Signals and Systems

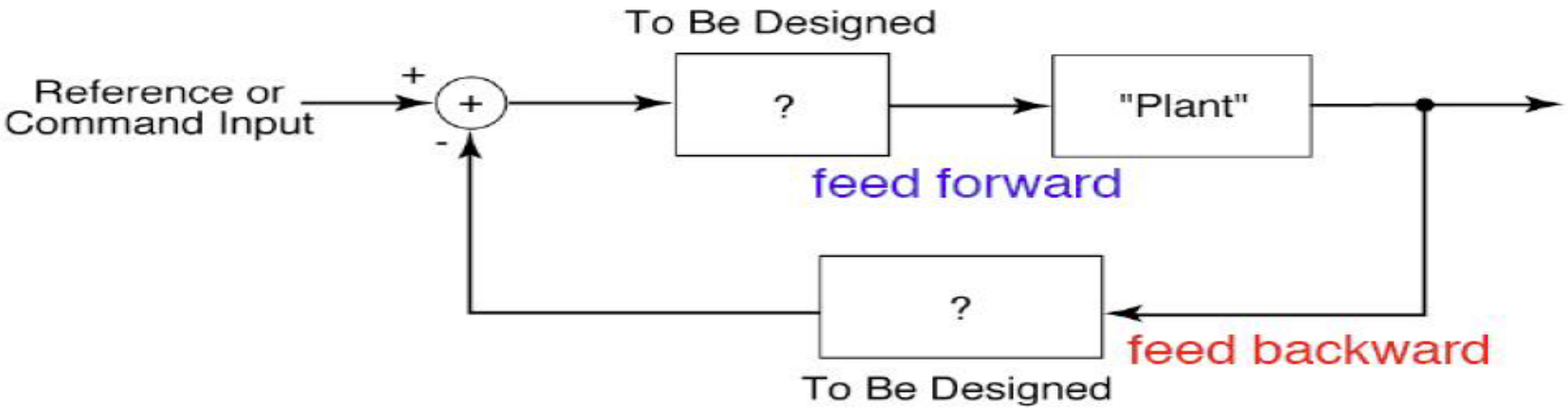
Lecture 09: CT Feedback and Control

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Zhejiang University

04/08/2025
Partly adapted from the materials provided on
the MIT OpenCourseWare

Review

A Typical Feedback System



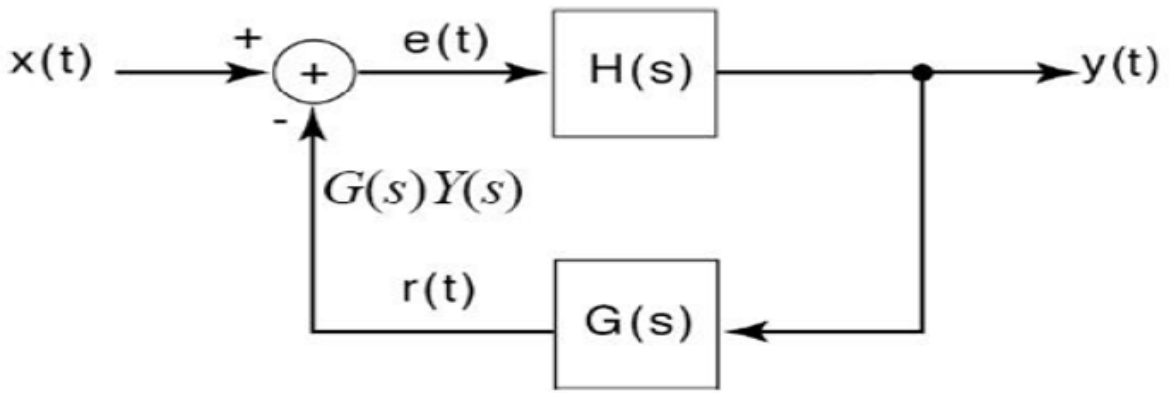
Why use Feedback?

- Reducing Nonlinearities
- Reducing Sensitivity to Uncertainties and Variability
- Stabilizing Unstable Systems
- Reducing Effects of Disturbances
- Tracking
- Shaping System Response Characteristics (bandwidth/speed)

:

System Function of a Closed-loop System

Example: A basic Feedback System — By its nature, we are dealing with real physical systems. ⇒ They are all *causal*.



$$E(s) = X(s) - R(s) = X(s) - G(s)Y(s)$$

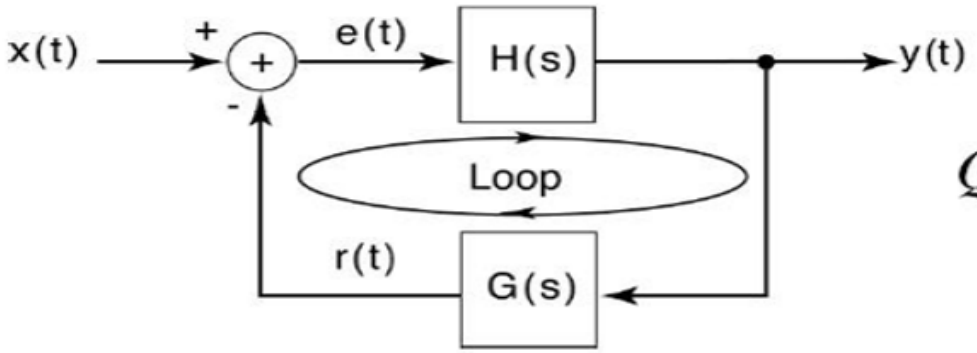
$$Y(s) = H(s)E(S) = H(s)[X(s) - G(s)Y(s)]$$

$$\Downarrow$$

$$Q(s) = \frac{Y(s)}{X(s)} = \frac{H(s)}{1 + G(s)H(s)}$$ — System function of the close-loop

General formula for a closed-loop system:

Black's Formula



$$Q(s) = \frac{Y(s)}{X(s)} = \frac{H(s)}{1 + G(s)H(s)}$$

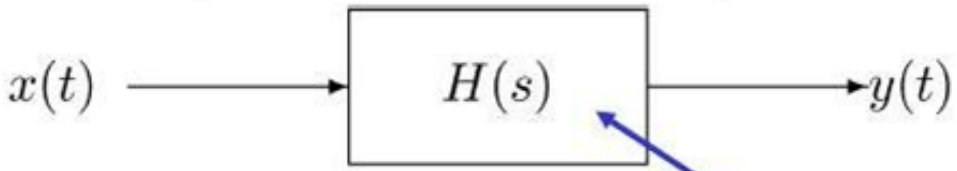
Can show for any closed-loop systems, the system function is given by Black's formula (H. S. Black in the 1920's, along with Nyquist and Bode):

$$\text{Closed - loop system function} = \frac{\text{forward gain}}{1 \text{ - loop gain}}$$

Forward gain — total gain along the forward path from the *input* to the *output*
the gain of an adder is ≡ 1

Loop gain — total gain along the closed loop — shared by all the system functions

Review



H(s) = "system function"

$$Y(s) = H(s)X(s)$$

1) System is stable $\Leftrightarrow \int_{-\infty}^{\infty} |h(t)|dt < \infty$ $\Leftrightarrow$ ROC of $H(s)$ includes $j\omega$ axis

2) Causality $\Rightarrow$ $h(t)$ right-sided signal $\Rightarrow$ ROC of $H(s)$ is a right-half plane

Question:

If the ROC of $H(s)$ is a right-half plane, is the system causal?

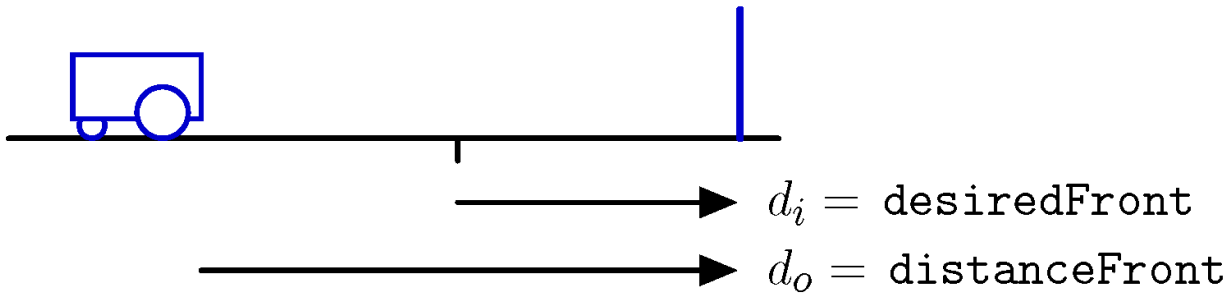
Ex. $H(s) = \dfrac{e^{sT}}{s+1}, \quad \Re e\{s\} > -1 \Rightarrow h(t)$ right-sided

$$h(t) = \mathcal{L}^{-1}\left\{\frac{e^{sT}}{s+1}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}_{t\to t+T} = e^{-t}u(t)|_{t\to t+T}$$

$$= e^{-(t+T)}u(t+T) \neq 0 \quad \text{at} \quad t < 0$$ Non-causal

Analysis of wallFinder System: Block Diagram

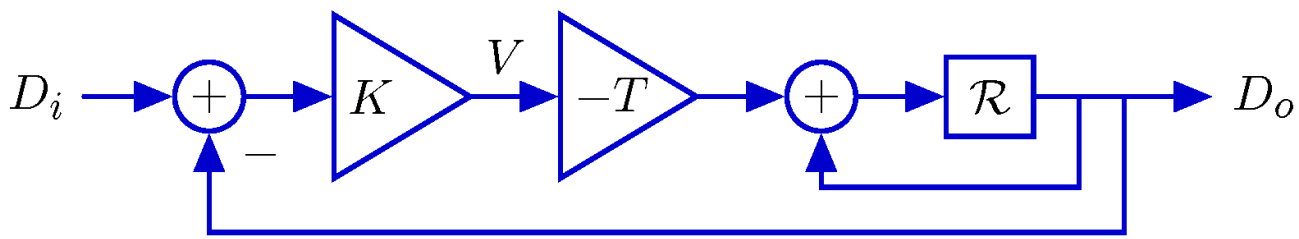
Visualize as block diagram.



proportional controller: $v[n] = Ke[n] = K\big(d_i[n] - d_s[n]\big)$

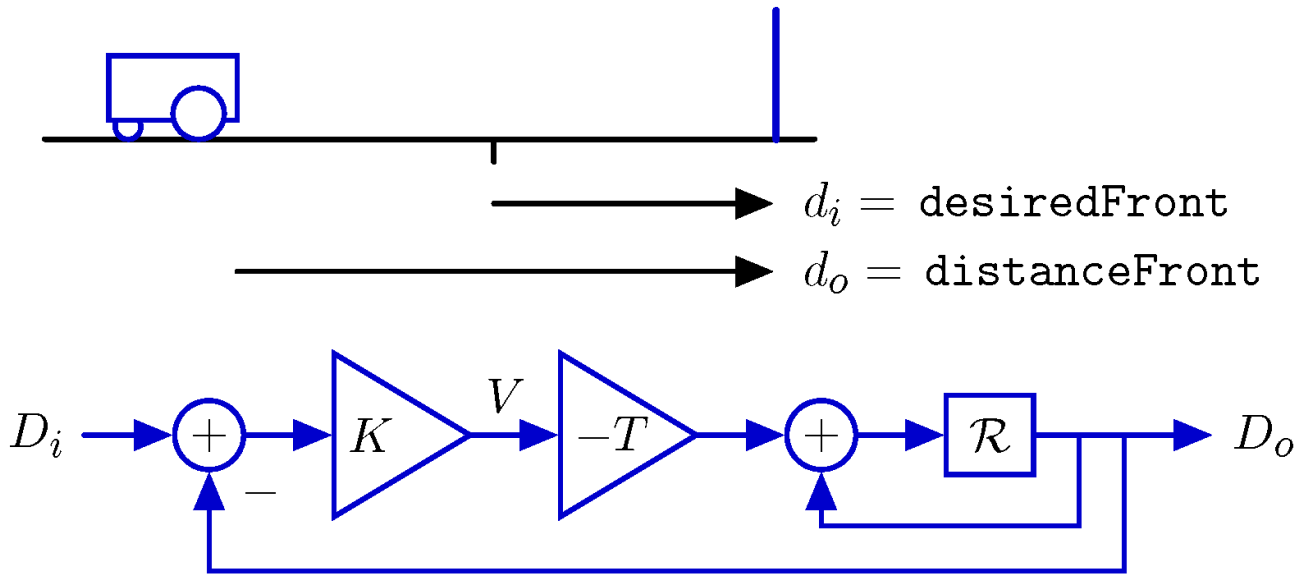
locomotion: $d_o[n] = d_o[n-1] - Tv[n-1]$

sensor with no delay: $d_s[n] = d_o[n]$



Analysis of wallFinder System: System Function

Solve.

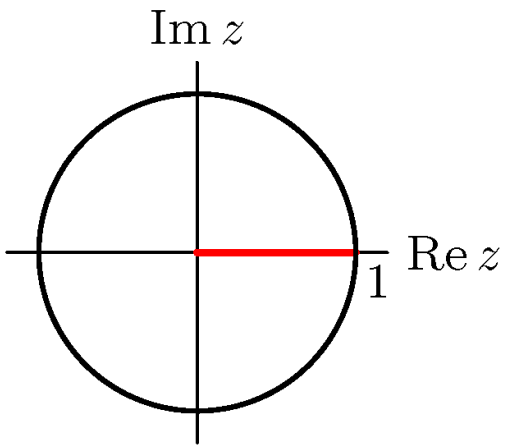


$$\frac{D_o}{D_i} = \frac{\frac{-KT\mathcal{R}}{1-\mathcal{R}}}{1+\frac{-KT\mathcal{R}}{1-\mathcal{R}}} = \frac{-KT\mathcal{R}}{1-\mathcal{R}-KT\mathcal{R}} = \frac{-KT\mathcal{R}}{1-(1+KT)\mathcal{R}}$$

Analysis of wallFinder System: Poles

The poles of the system function provide insight for choosing K.

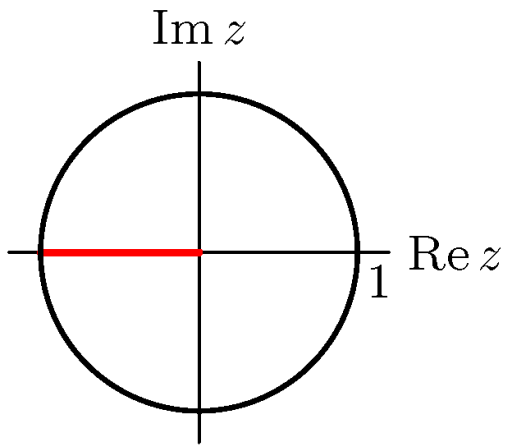
$$\frac{D_o}{D_i} = \frac{-KT\mathcal{R}}{1-(1+KT)\mathcal{R}} = \frac{(1-p_o)\mathcal{R}}{1-p_o\mathcal{R}} \; ; \quad p_0 = 1+KT$$



$0 < p_0 < 1$

$-1 < KT < 0$

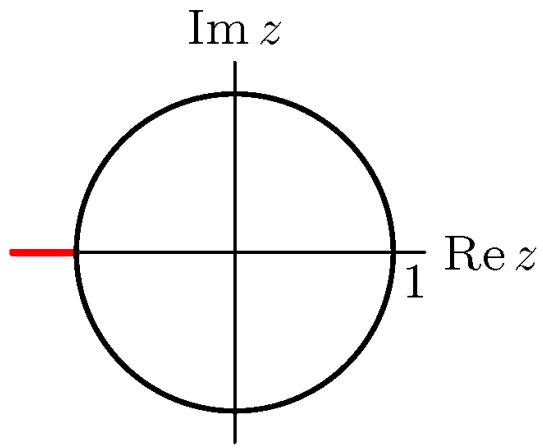
monotonic converging



$-1 < p_0 < 0$

$-2 < KT < -1$

alternating converging



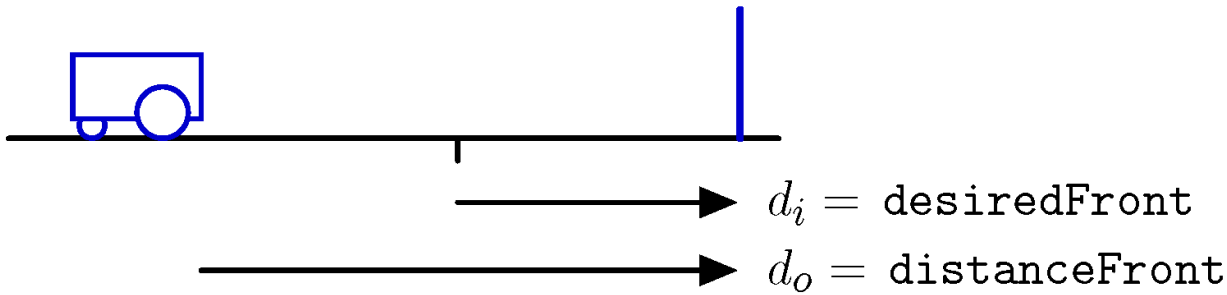
$p_0 < -1$

$KT < -2$

alternating diverging

Analysis of wallFinder System: Adding Sensor Delay

Adding delay tends to destabilize control systems.



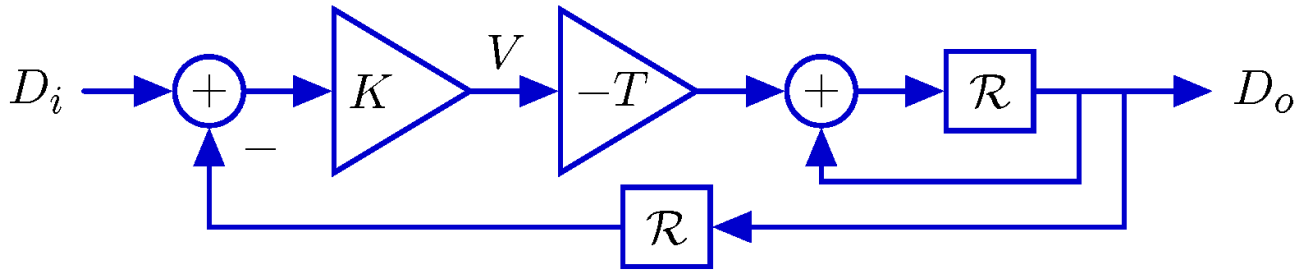
proportional controller: $v[n] = Ke[n] = K\big(d_i[n] - d_s[n]\big)$

locomotion: $d_o[n] = d_o[n-1] - Tv[n-1]$

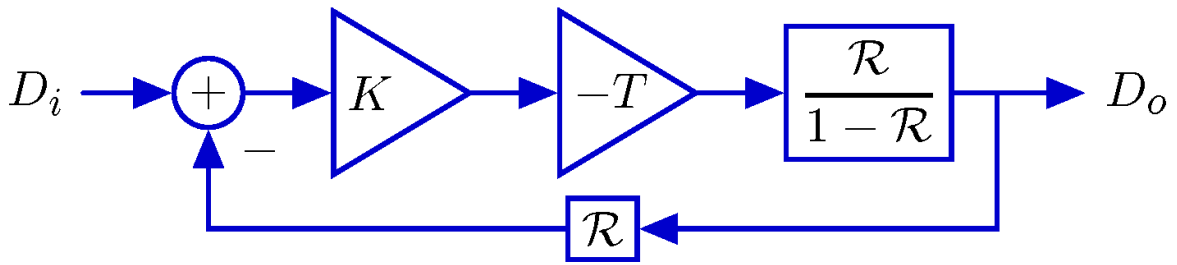
sensor **with delay**: $d_s[n] = d_o[\mathbf{n-1}]$

Check Yourself

Find the system function $H = \dfrac{D_o}{D_i}$.



Replace accumulator with equivalent block diagram.



$$\frac{D_o}{D_i} = \frac{\dfrac{-KT\mathcal{R}}{1-\mathcal{R}}}{1 + \dfrac{-KT\mathcal{R}^2}{1-\mathcal{R}}} = \frac{-KT\mathcal{R}}{1-\mathcal{R}-KT\mathcal{R}^2}$$

Analyzing wallFinder: Poles

Substitute $\mathcal{R} \to \frac{1}{z}$ in the system functional to find the poles.

$$\frac{D_o}{D_i} = \frac{-KT\mathcal{R}}{1 - \mathcal{R} - KT\mathcal{R}^2} = \frac{-KT\frac{1}{z}}{1 - \frac{1}{z} - KT\frac{1}{z^2}} = \frac{-KTz}{z^2 - z - KT}$$

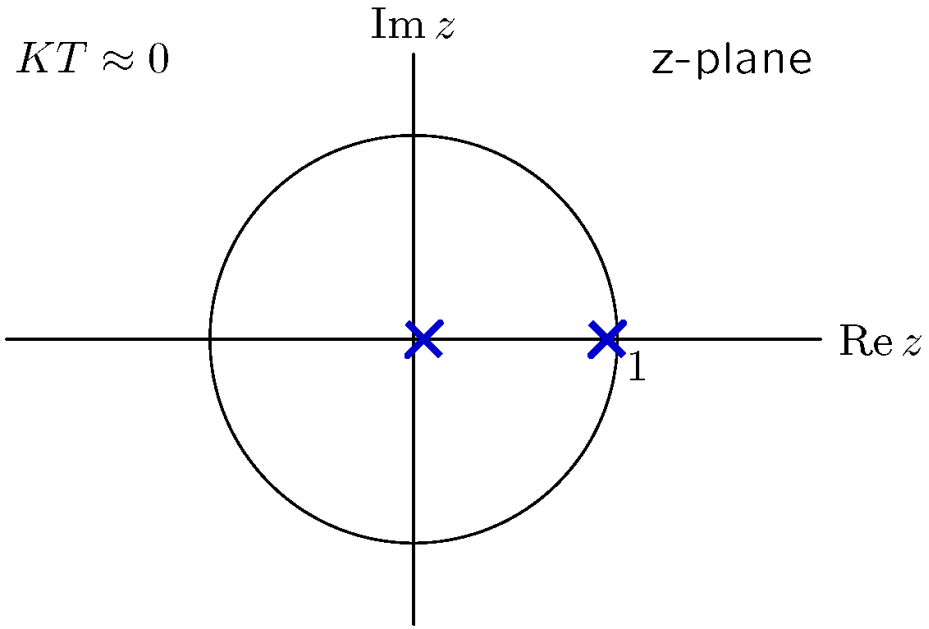
The poles are then the roots of the denominator.

$$z = \frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2 + KT}$$

Feedback and Control: Poles

If KT is small, the poles are at $z \approx -KT$ and $z \approx 1 + KT$.

$$z = \tfrac{1}{2} \pm \sqrt{\left(\tfrac{1}{2}\right)^2 + KT} \approx \tfrac{1}{2} \pm \sqrt{\left(\tfrac{1}{2} + KT\right)^2} = 1 + KT,\ -KT$$



Pole near 0 generates fast response.

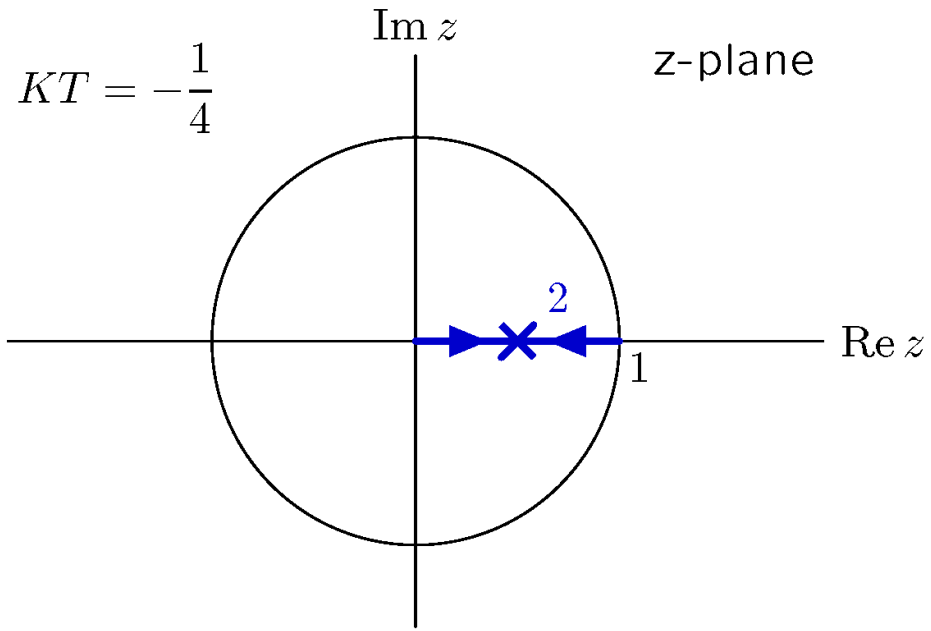
Pole near 1 generates slow response.

Slow mode (pole near 1) dominates the response.

Feedback and Control: Poles

As KT becomes more negative, the poles move toward each other and collide at $z = \frac{1}{2}$ when $KT = -\frac{1}{4}$.

$$z = \frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2 + KT} = \frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2 - \frac{1}{4}} = \frac{1}{2}, \frac{1}{2}$$

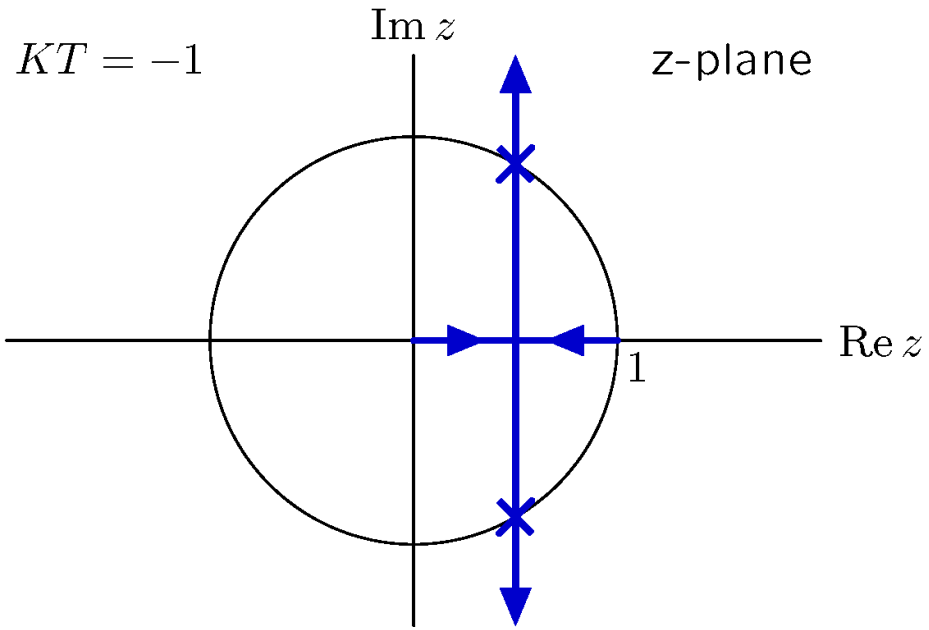


Persistent responses decay. The system is stable.

Feedback and Control: Poles

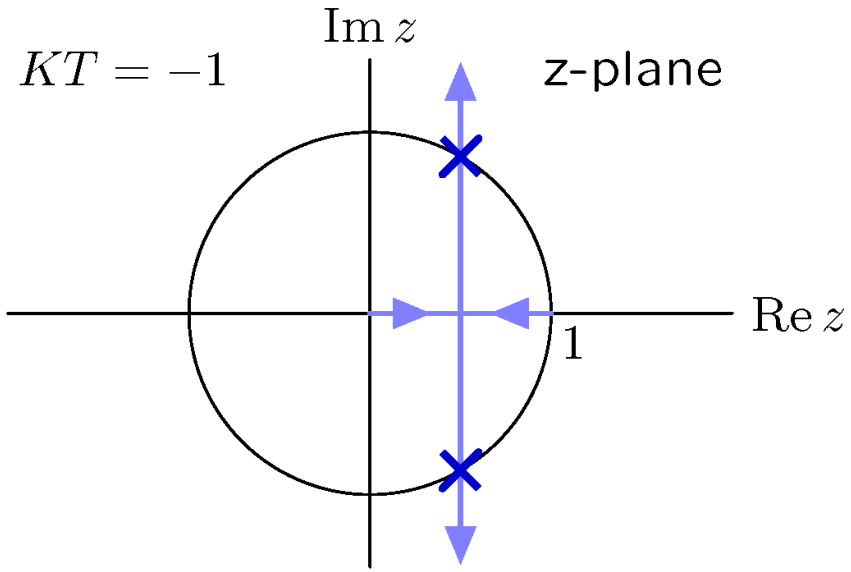
If $KT < -1/4$, the poles are complex.

$$z = \tfrac{1}{2} \pm \sqrt{\left(\tfrac{1}{2}\right)^2 + KT} = \tfrac{1}{2} \pm j\sqrt{-KT - \left(\tfrac{1}{2}\right)^2}$$



Complex poles $\rightarrow$ oscillations.

Check Yourself



$$p_0 = \frac{1}{2} \pm j\frac{\sqrt{3}}{2} = e^{\pm j\pi/3}$$

$$p_0^n = e^{\pm j\pi n/3}$$

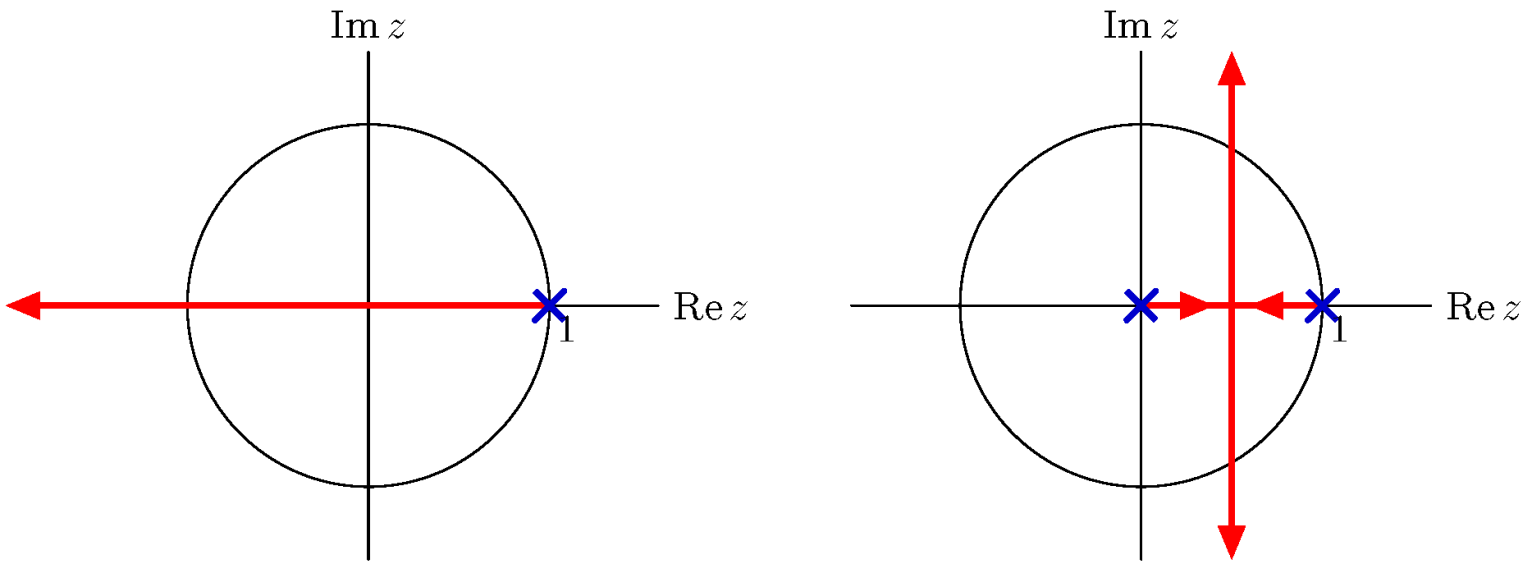
$$\underbrace{e^{\pm j0\pi/3}}_{1},\ e^{\pm j\pi/3},\ e^{\pm j2\pi/3},\ e^{\pm j3\pi/3},\ e^{\pm j4\pi/3},\ e^{\pm j5\pi/3},\ \underbrace{e^{\pm j6\pi/3}}_{e^{\pm j2\pi}=1}$$

Destabilizing Effect of Delay

Adding delay in the feedback loop makes it more difficult to stabilize.

Ideal sensor: $d_s[n] = d_o[n]$

More realistic sensor (with delay): $d_s[n] = d_o[n-1]$

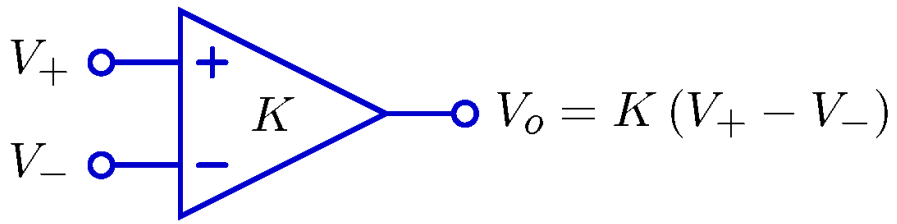


Fastest response without delay: single pole at $z = 0$.

Fastest response with delay: double pole at $z = \frac{1}{2}$. **much slower!**

Op-Amp

An "ideal" op-amp has many desireable characteristics.



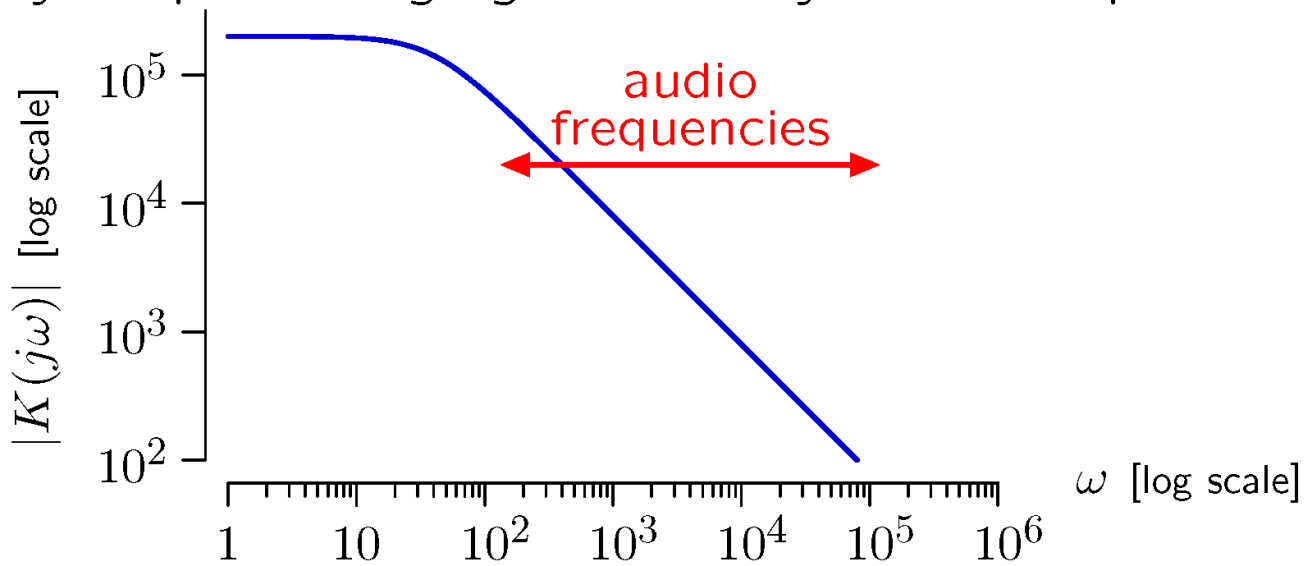
- high speed
- large bandwidth
- high input impedance
- low output impedance
- ...

It is difficult to build a circuit with all of these features.

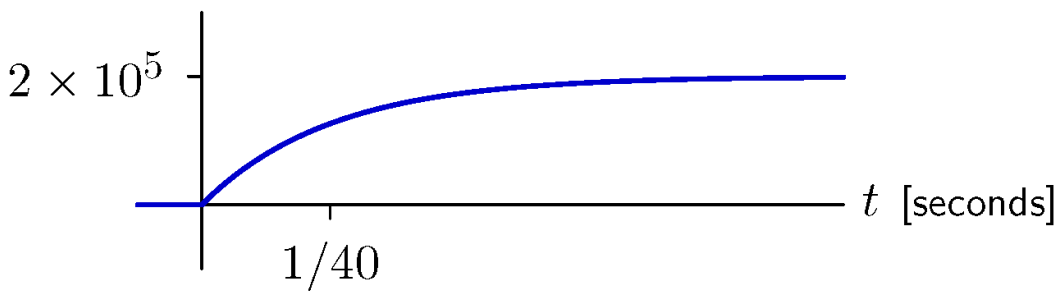
Op-Amp

Performance parameters for real op-amps fall short of the ideal.

Frequency Response: high gain but only at low frequencies.



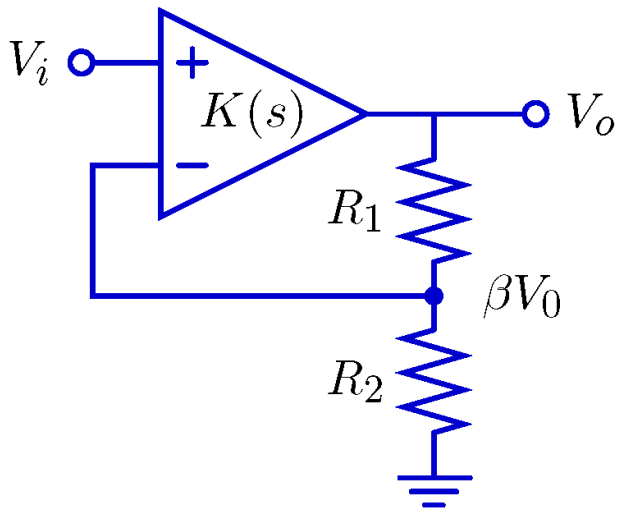
Step Response: slow by electronic standards.



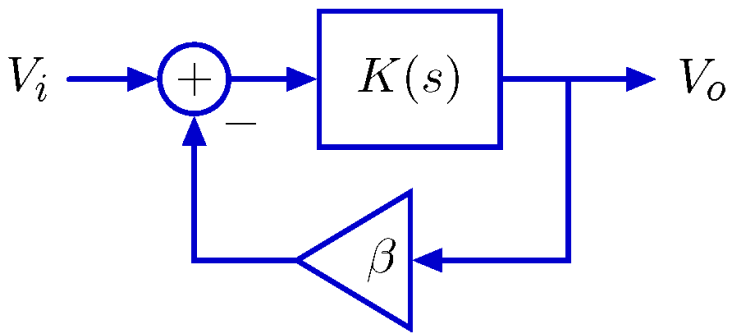
Improving Performance

Using feedback to improve performance parameters.

circuit



6.003 model

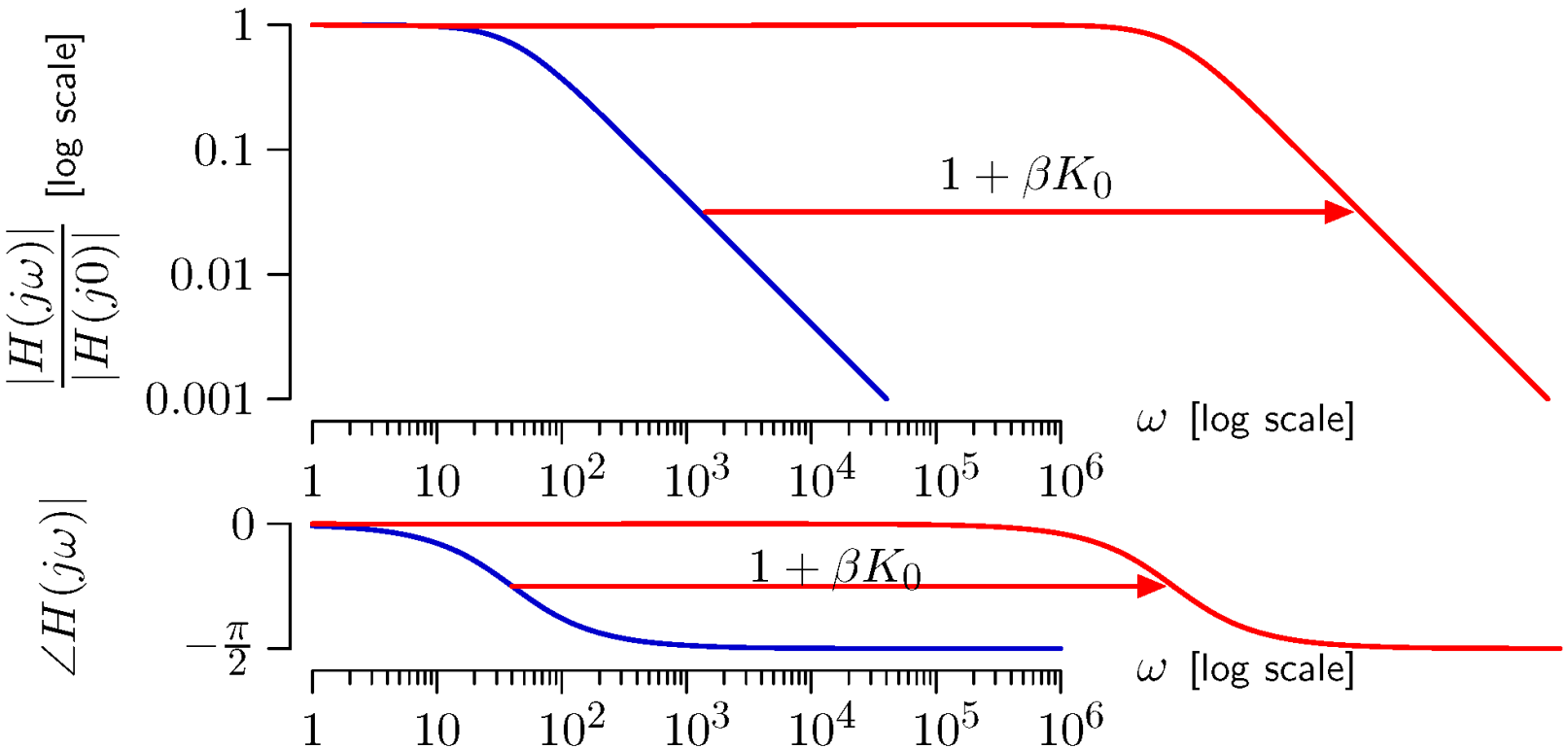


$$V_- = \beta V_o = \left(\frac{R_2}{R_1 + R_2}\right) V_o$$

$$\begin{aligned}\frac{V_o}{V_i} &= \frac{K(s)}{1 + \beta K(s)} \\ &= \frac{\frac{\alpha K_0}{s+\alpha}}{1 + \beta\frac{\alpha K_0}{s+\alpha}} \\ &= \frac{\alpha K_0}{s + \alpha + \alpha\beta K_0}\end{aligned}$$

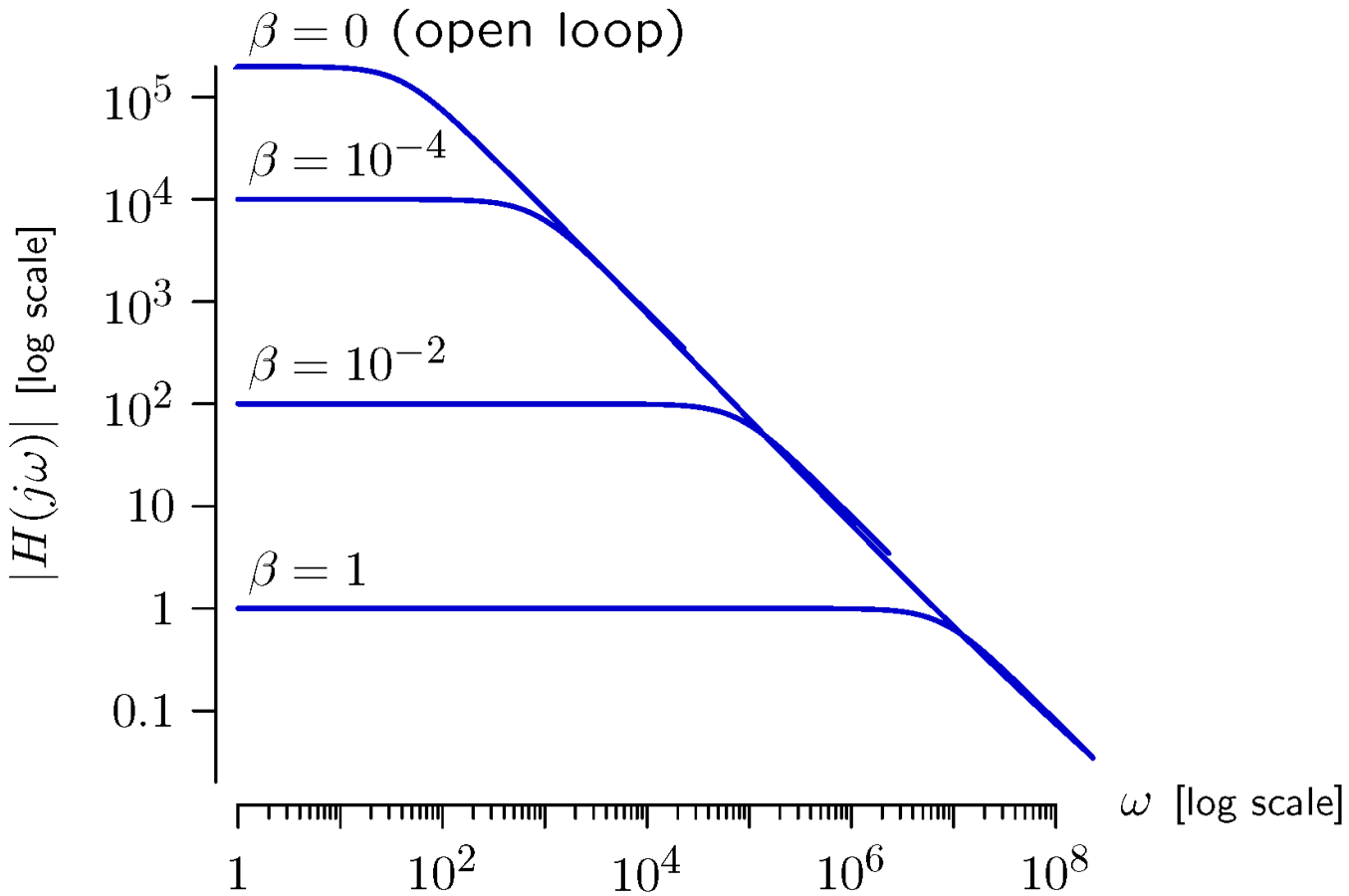
Improving Performance

Feedback extends frequency response by a factor of $1 + \beta K_0$ ($K_0 = 2 \times 10^5$).



Improving Performance

Feedback produces higher bandwidths by **reducing** the gain at low frequencies. It trades gain for bandwidth.

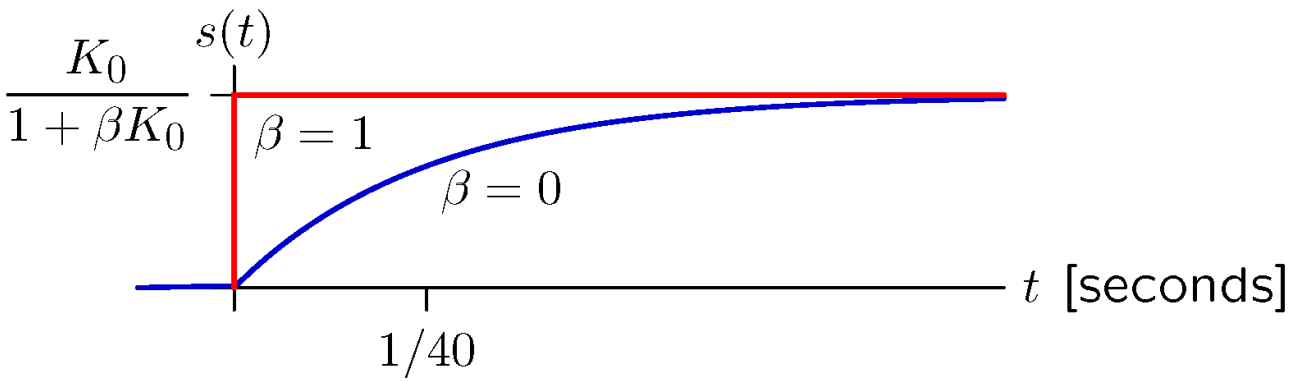


Improving Performance

Feedback makes the time response faster by a factor of $1 + \beta K_0$ ($K_0 = 2 \times 10^5$).

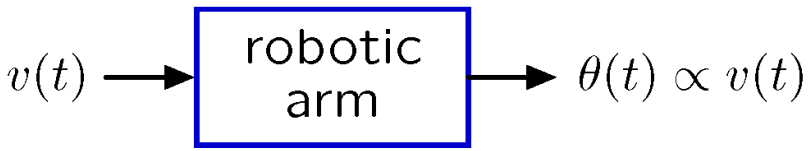
Step response

$$s(t) = \frac{K_0}{1 + \beta K_0}(1 - e^{-\alpha(1+\beta K_0)t})u(t)$$

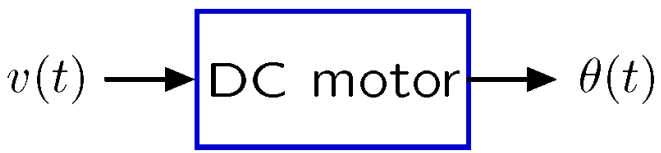


Motor Controller

We wish to build a robot arm (actually its elbow). The input should be voltage $v(t)$, and the output should be the elbow angle $\theta(t)$.



We wish to build the robot arm with a DC motor.



This problem is similar to the head-turning servo in 6.01 !

Check Yourself

What is the relation between $v(t)$ and $\theta(t)$ for a DC motor?

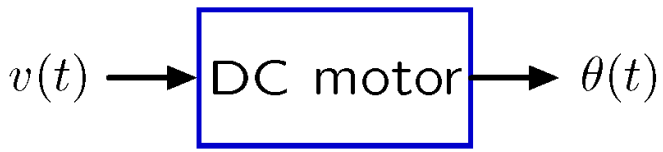


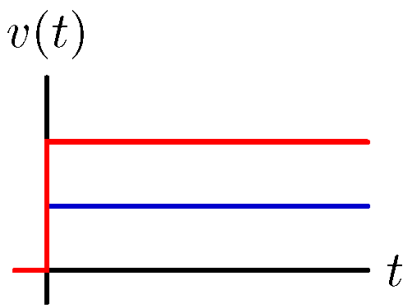
1. $\theta(t) \propto v(t)$
2. $\cos\theta(t) \propto v(t)$
3. $\theta(t) \propto \dot{v}(t)$
4. $\cos\theta(t) \propto \dot{v}(t)$
5. none of the above

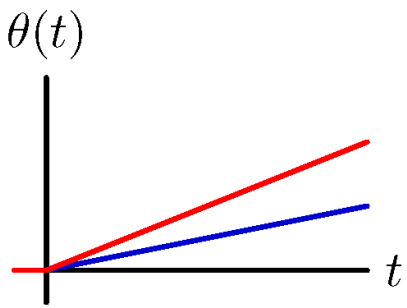
Check Yourself

What is the relation between $v(t)$ and $\theta(t)$ for a DC motor?

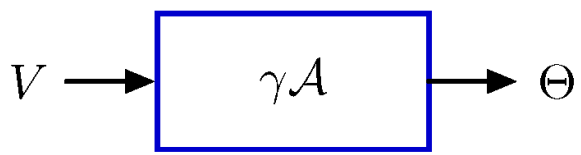
To first order, the rotational speed $\dot{\theta}(t)$ of a DC motor is proportional to the input voltage $v(t)$.







First-order model: integrator



Check Yourself

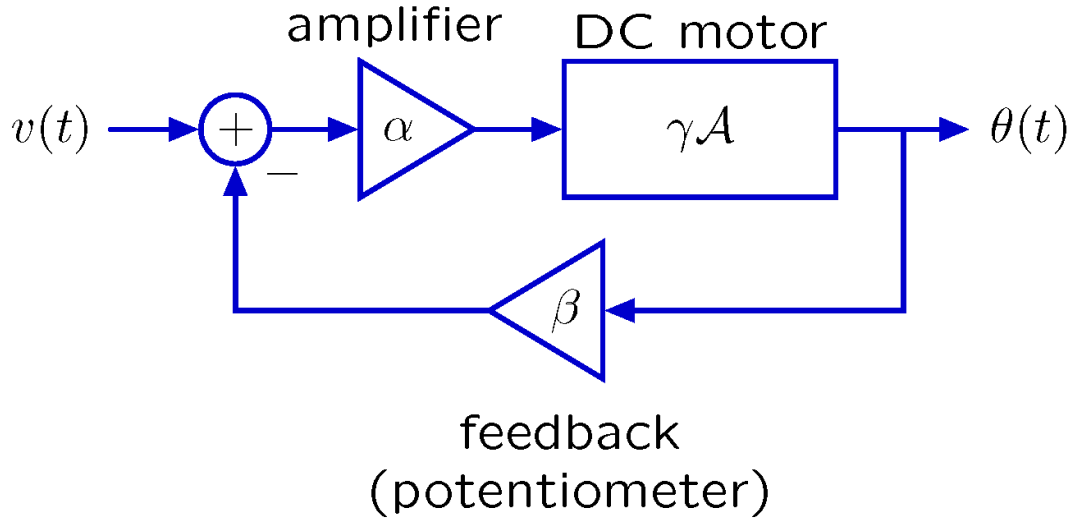
What is the relation between $v(t)$ and $\theta(t)$ for a DC motor?



1. $\theta(t) \propto v(t)$
2. $\cos\theta(t) \propto v(t)$
3. $\theta(t) \propto \dot{v}(t)$
4. $\cos\theta(t) \propto \dot{v}(t)$
5. none of the above

Motor Controller

Use proportional feedback to control the angle of the motor's shaft.

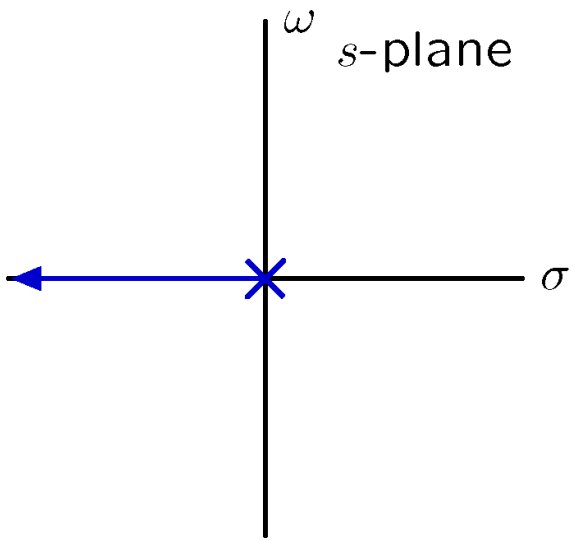


$$\frac{\Theta}{V} = \frac{\alpha\gamma\mathcal{A}}{1+\alpha\beta\gamma\mathcal{A}} = \frac{\alpha\gamma\frac{1}{s}}{1+\alpha\beta\gamma\frac{1}{s}} = \frac{\alpha\gamma}{s+\alpha\beta\gamma}$$

Motor Controller

The closed loop system has a single pole at $s = -\alpha\beta\gamma$.

$$\frac{\Theta}{V} = \frac{\alpha\gamma}{s + \alpha\beta\gamma}$$



As α increases, the closed-loop pole becomes increasingly negative.

Motor Controller

Find the impulse and step response.

The system function is

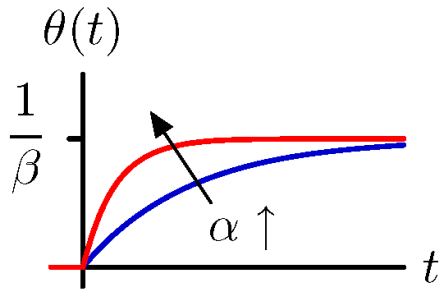
$$\frac{\Theta}{V} = \frac{\alpha\gamma}{s + \alpha\beta\gamma}\,.$$

The impulse response is

$$h(t) = \alpha\gamma e^{-\alpha\beta\gamma t} u(t)$$

and the step response is therefore

$$s(t) = \frac{1}{\beta}\left(1 - e^{-\alpha\beta\gamma t}\right) u(t)\,.$$



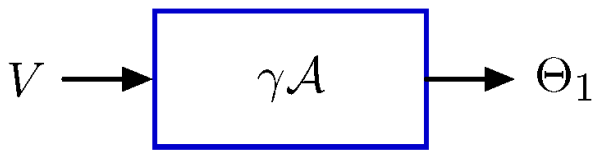
The response is faster for larger values of α.

Try it: Demo.

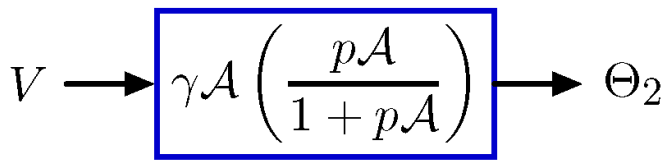
Motor Controller

The speed of a DC motor does not change instantly if the voltage is stepped. There is lag due to rotational inertia.

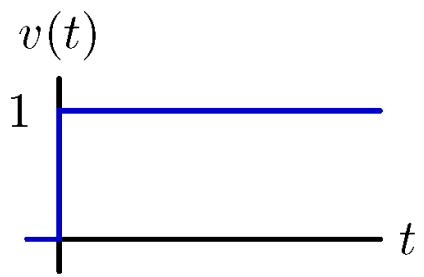
First-order model
integrator

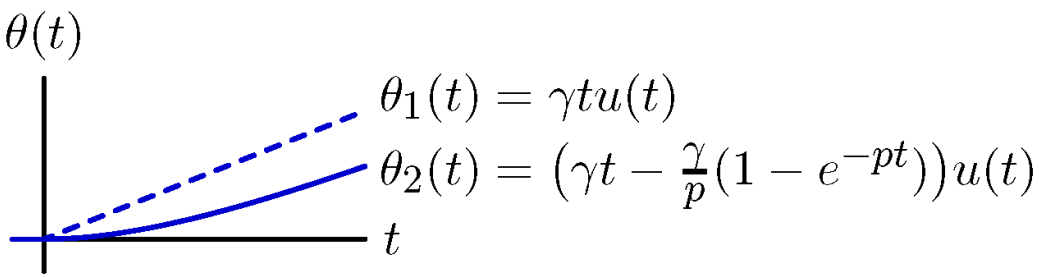


Second-order model
integrator with lag



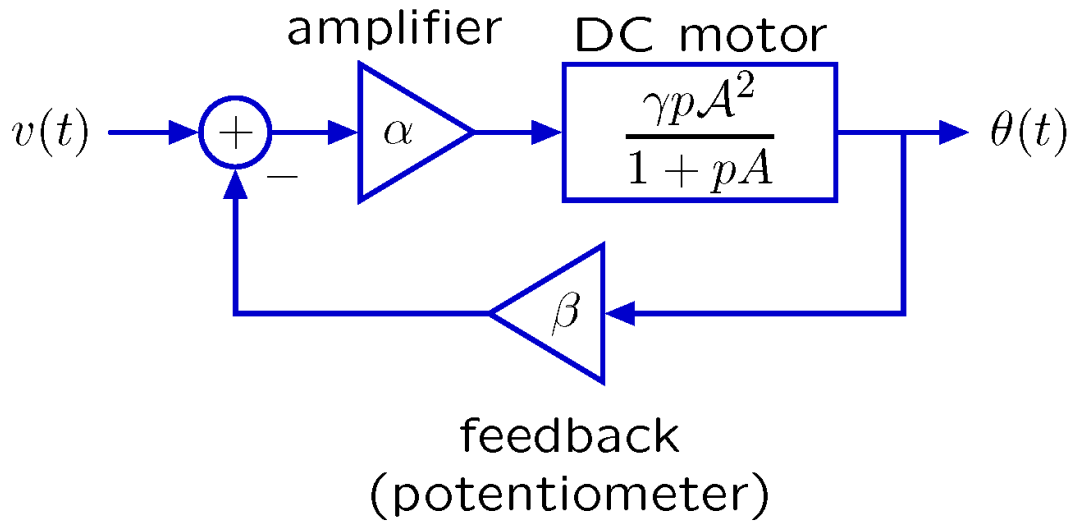
Step response of the models:





Motor Controller

Analyze second-order model.

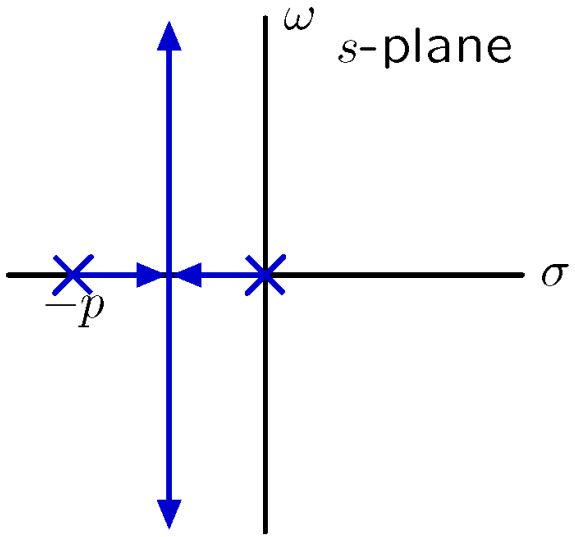


$$\frac{\Theta}{V} = \frac{\frac{\alpha\gamma p\mathcal{A}^2}{1+pA}}{1+\frac{\alpha\beta\gamma p\mathcal{A}^2}{1+pA}} = \frac{\alpha\gamma p\mathcal{A}^2}{1+pA+\alpha\beta\gamma p\mathcal{A}^2} = \frac{\alpha\gamma p}{s^2+ps+\alpha\beta\gamma p}$$

$$s = -\frac{p}{2} \pm \sqrt{\left(\frac{p}{2}\right)^2 - \alpha\beta\gamma p}$$

Motor Controller

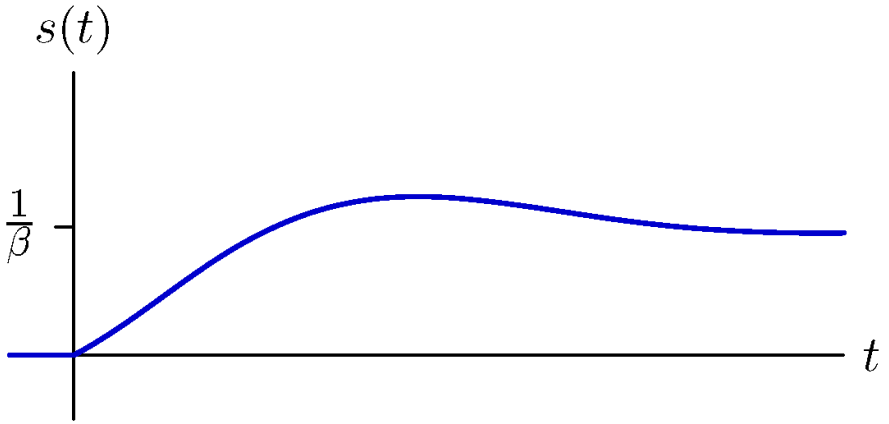
For second-order model, increasing α causes the poles at 0 and $-p$ to approach each other, collide at $s = -p/2$, then split into two poles with imaginary parts.



Increasing the gain α does not increase speed of convergence.

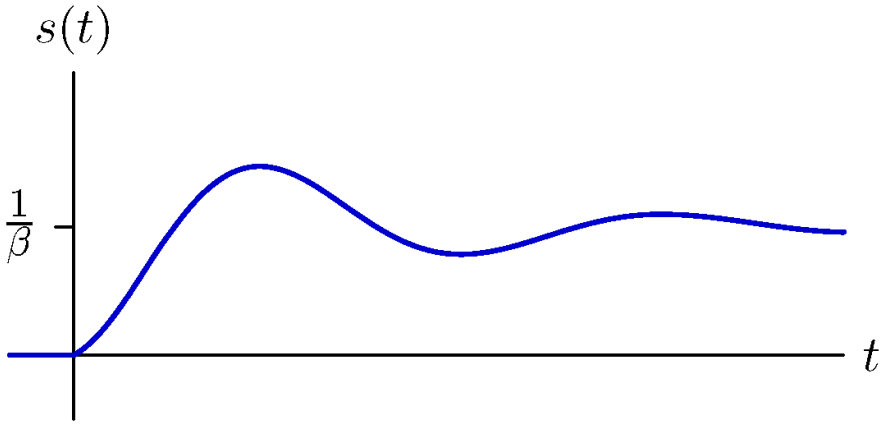
Motor Controller

Step response.



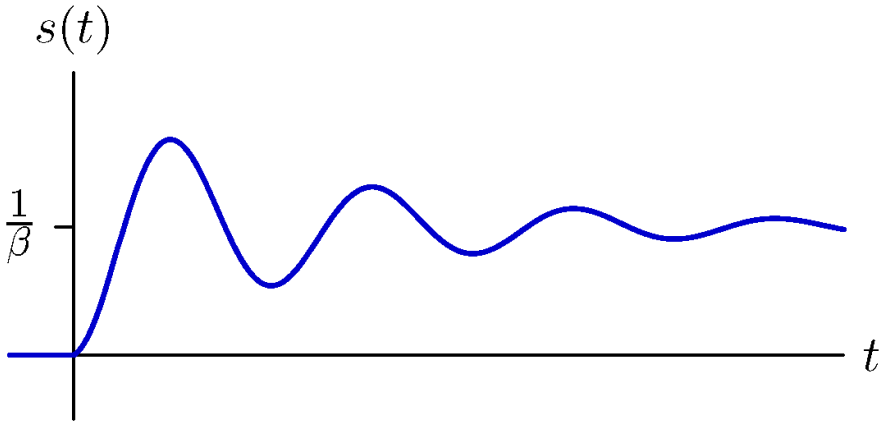
Motor Controller

Step response.



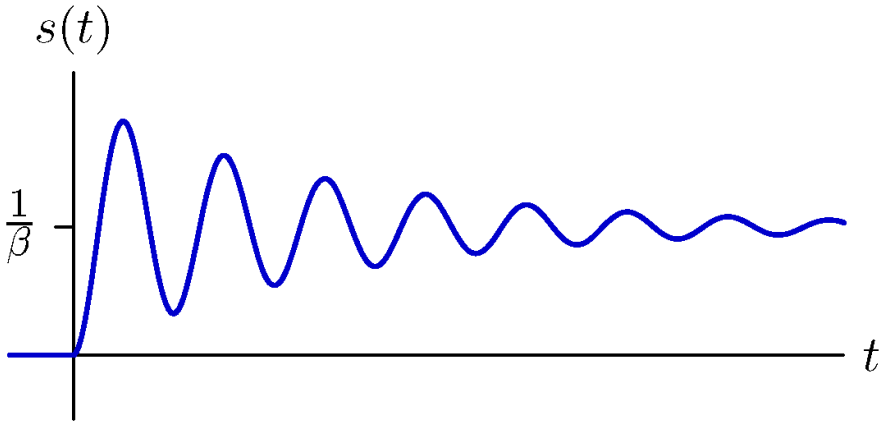
Motor Controller

Step response.



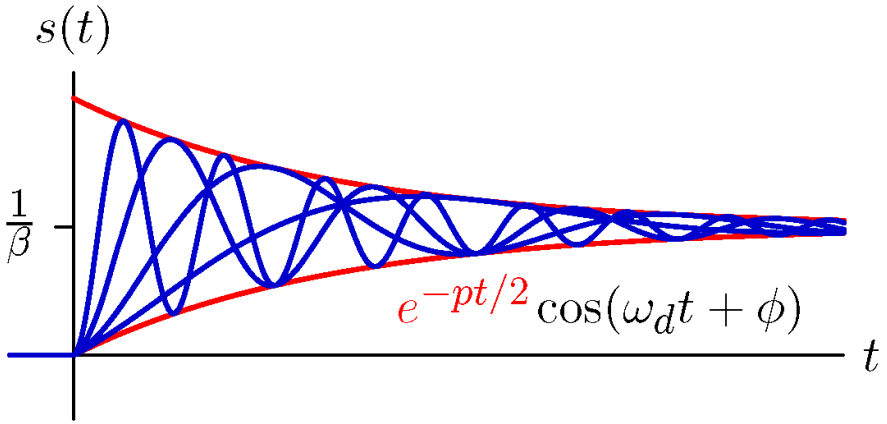
Motor Controller

Step response.



Motor Controller

Step response.



Feedback and Control

Using feedback to enhance performance.

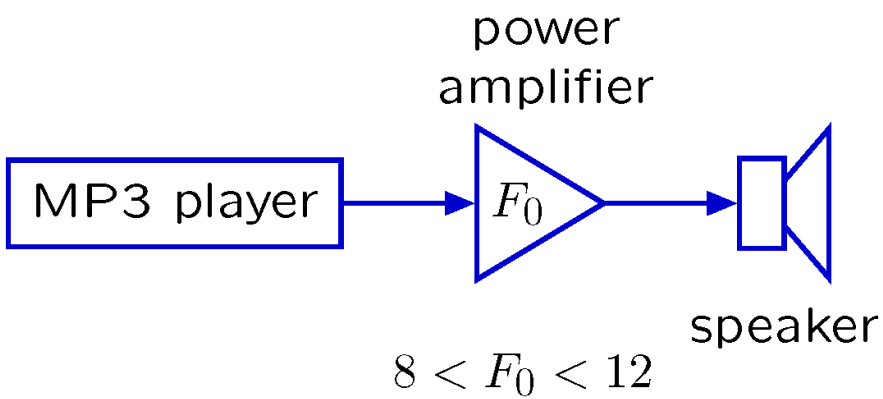
Examples:

- improve performance of an op amp circuit.
- control position of a motor.
- reduce sensitivity to unwanted parameter variation.
- reduce distortions.
- stabilize unstable systems
  - magnetic levitation
  - inverted pendulum

Feedback and Control

Reducing sensitivity to unwanted parameter variation.

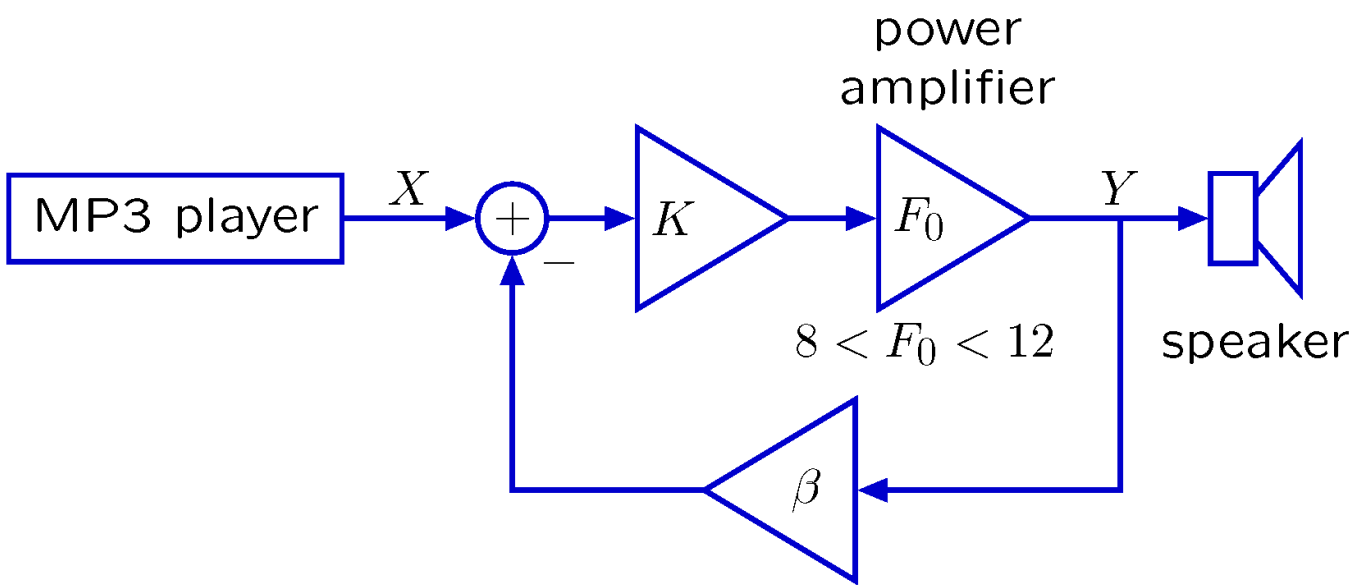
Example: power amplifier



Changes in F_0 (due to changes in temperature, for example) lead to undesired changes in sound level.

Feedback and Control

Feedback can be used to compensate for parameter variation.



$$H(s) = \frac{KF_0}{1 + \beta K F_0}$$

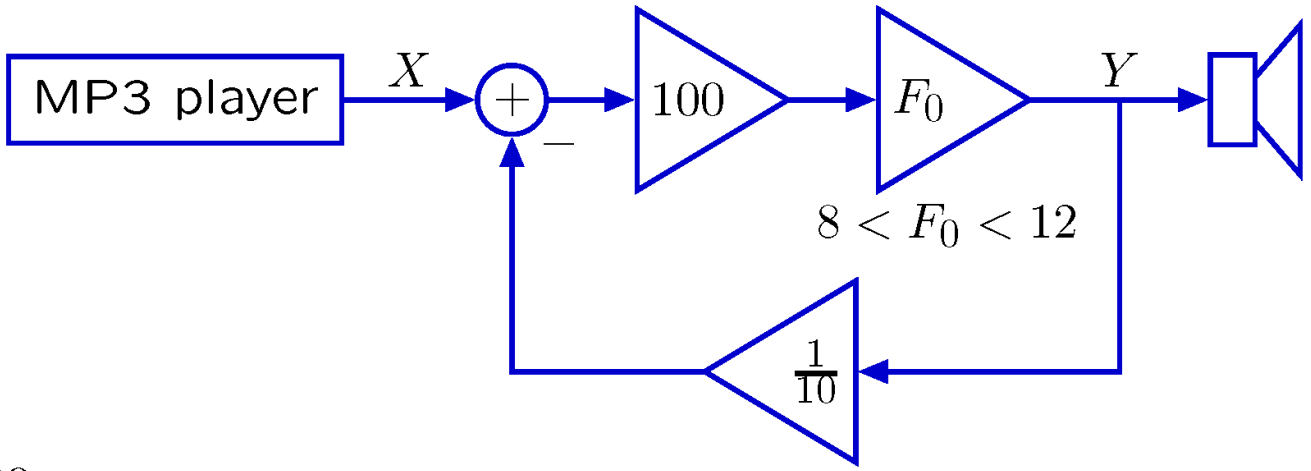
If K is made large, so that $\beta K F_0 \gg 1$, then

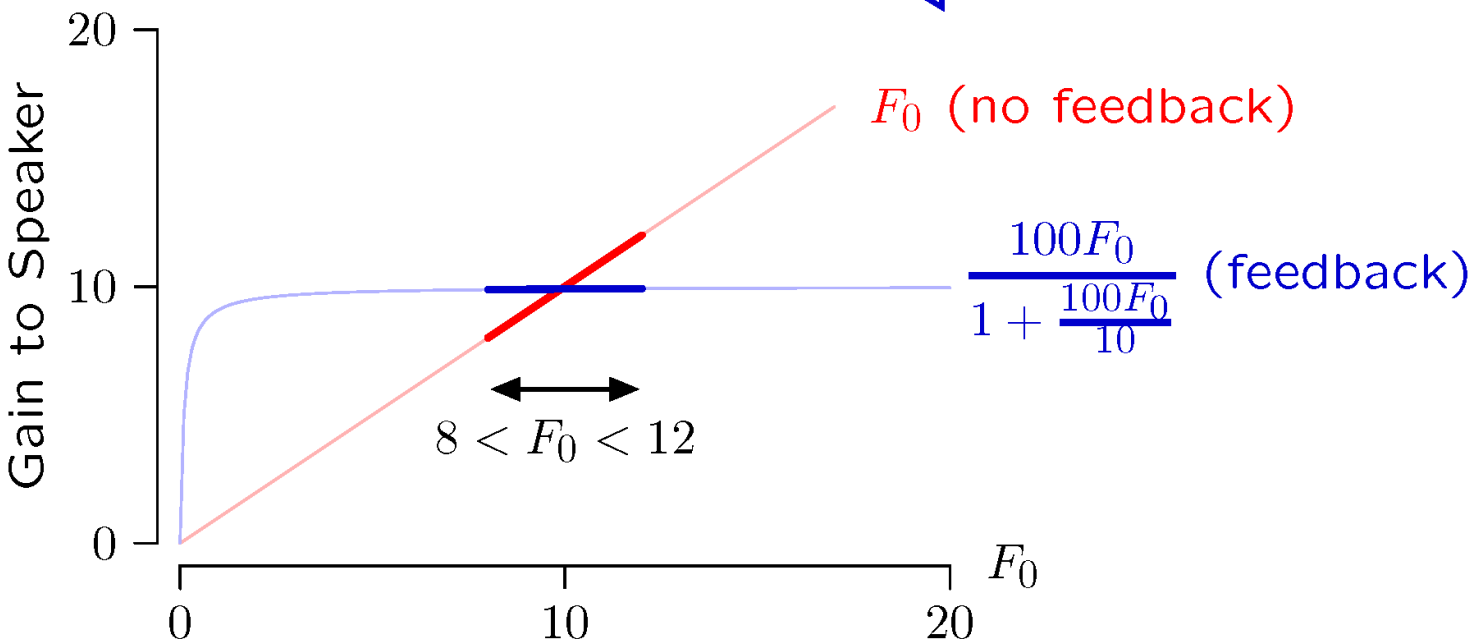
$$H(s) \approx \frac{1}{\beta}$$

independent of K or F_0!

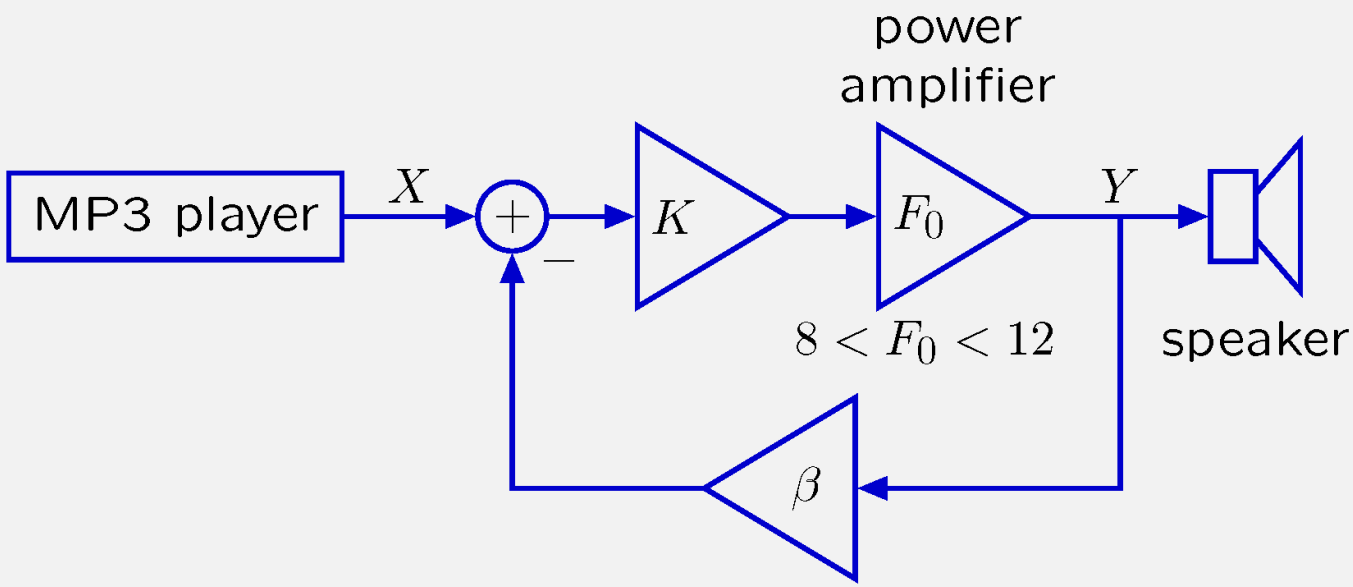
Feedback and Control

Feedback reduces the change in gain due to change in F_0.





Check Yourself



Feedback greatly reduces sensitivity to variations in K or F_0.

$$\lim_{K\to\infty} H(s) = \frac{KF_0}{1+\beta KF_0} \to \frac{1}{\beta}$$

What about variations in β? Aren't those important?

Check Yourself

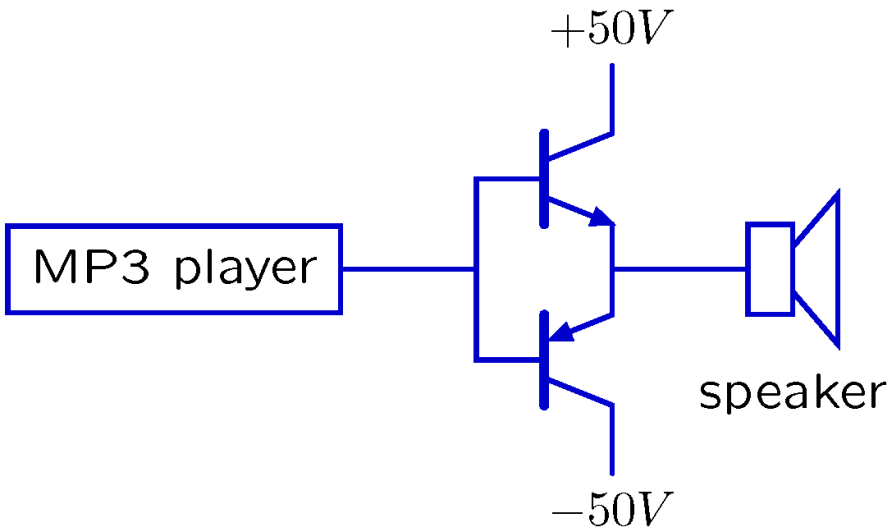
What about variations in β? Aren't those important?

The value of β is typically determined with resistors, whose values are quite stable (compared to semiconductor devices).

Crossover Distortion

Feedback can compensate for parameter variation even when the variation occurs rapidly.

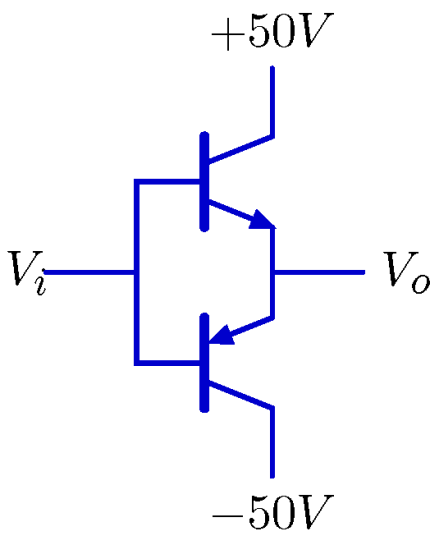
Example: using transistors to amplify power.

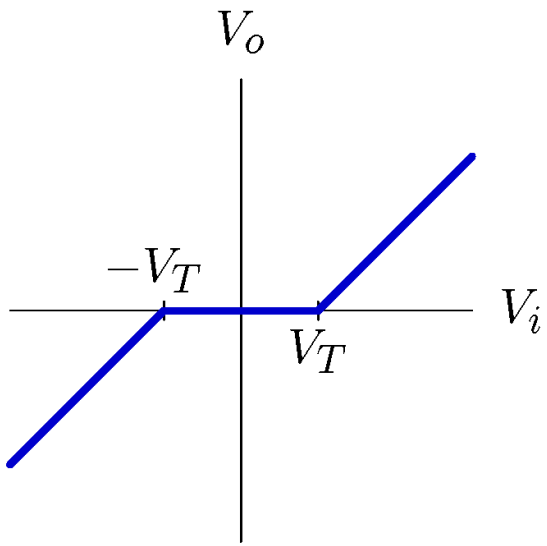


Crossover Distortion

This circuit introduces "crossover distortion."

For the upper transistor to conduct, $V_i - V_o > V_T$.
For the lower transistor to conduct, $V_i - V_o < -V_T$.

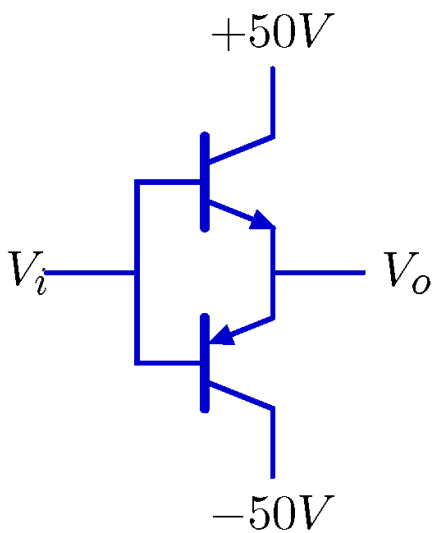


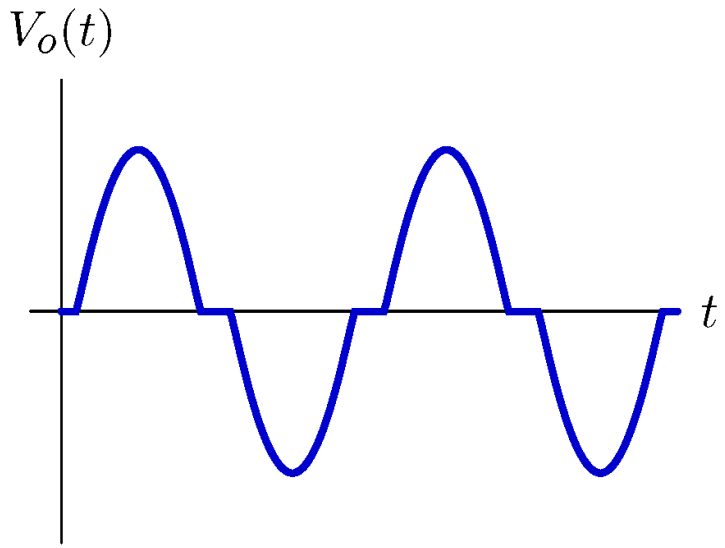


Crossover Distortion

Crossover distortion can have dramatic effects.

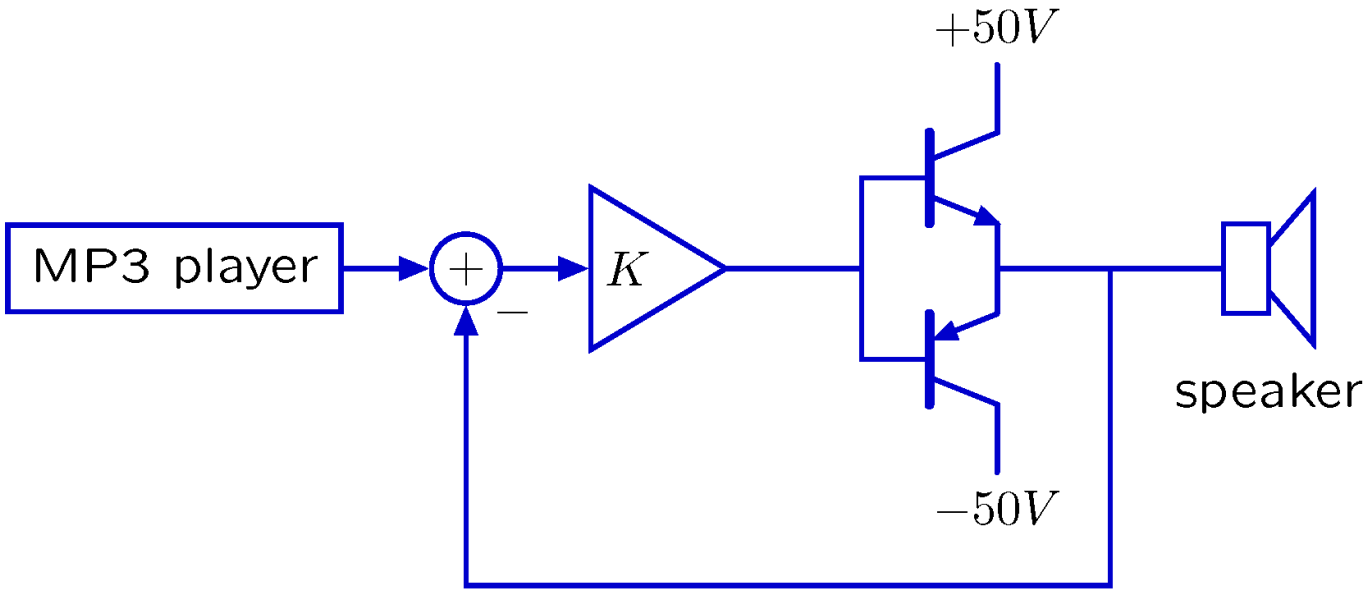
Example: crossover distortion when the input is $V_i(t) = B\sin(\omega_0 t)$.





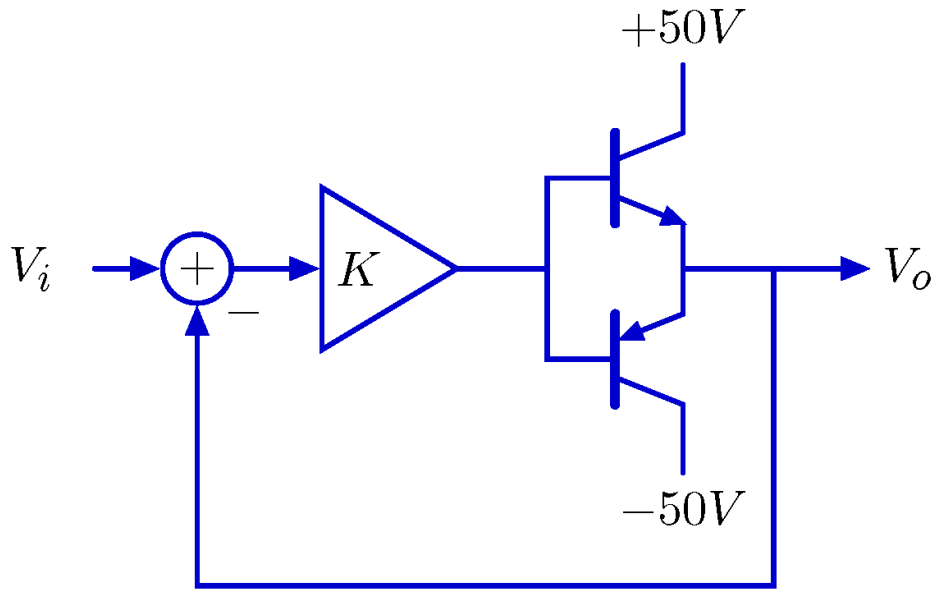
Crossover Distortion

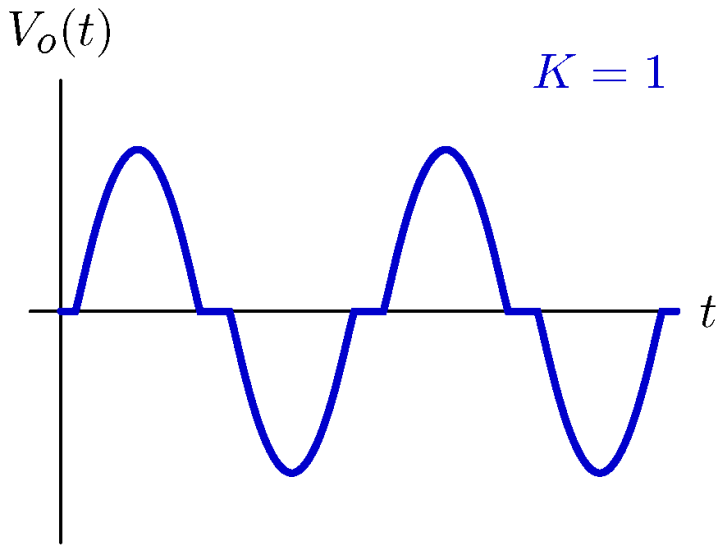
Feedback can reduce the effects of crossover distortion.



Crossover Distortion

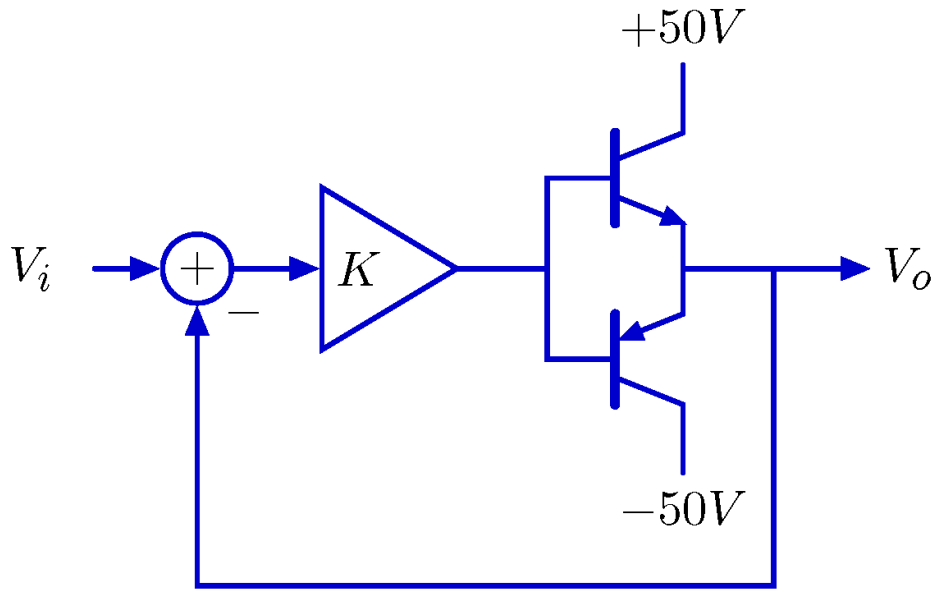
When K is small, feedback has little effect on crossover distortion.

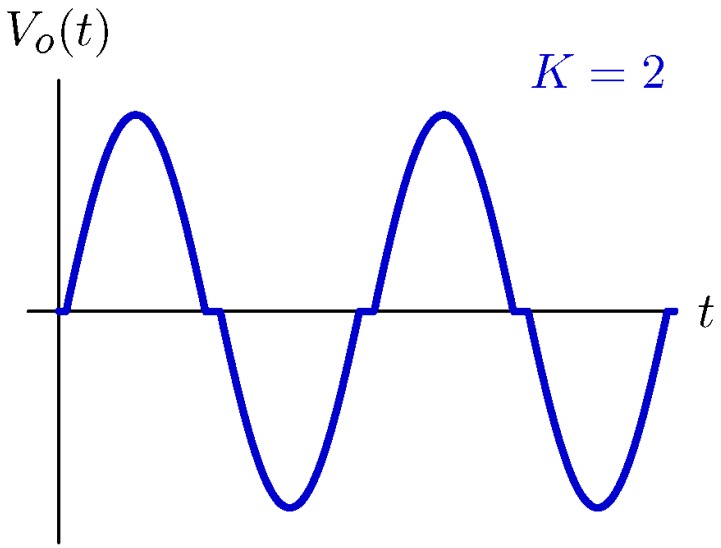




Crossover Distortion

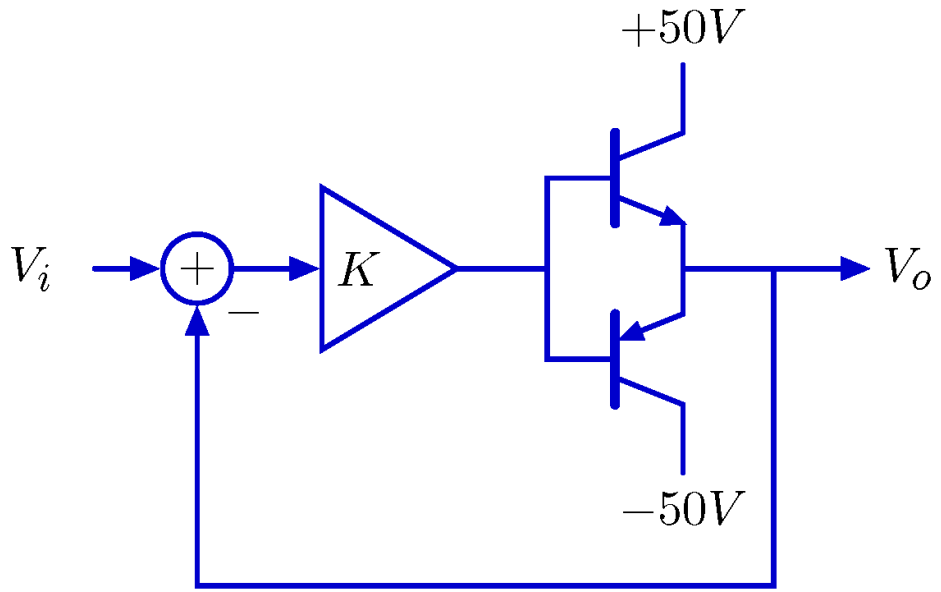
As K increases, feedback reduces crossover distortion.

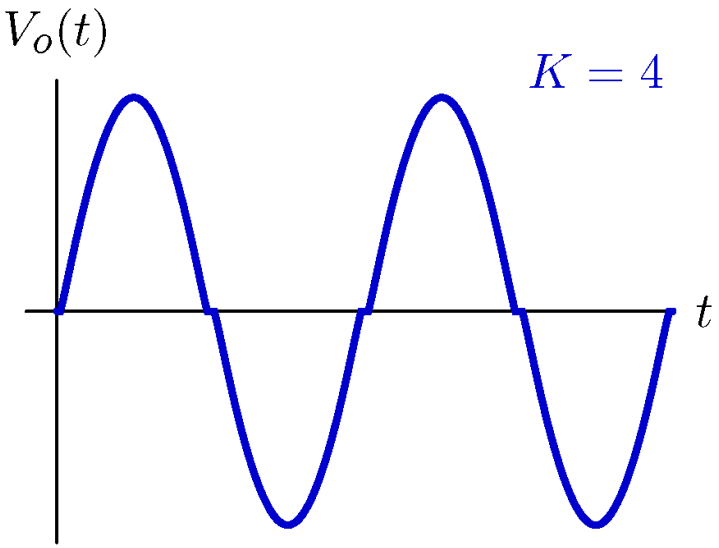




Crossover Distortion

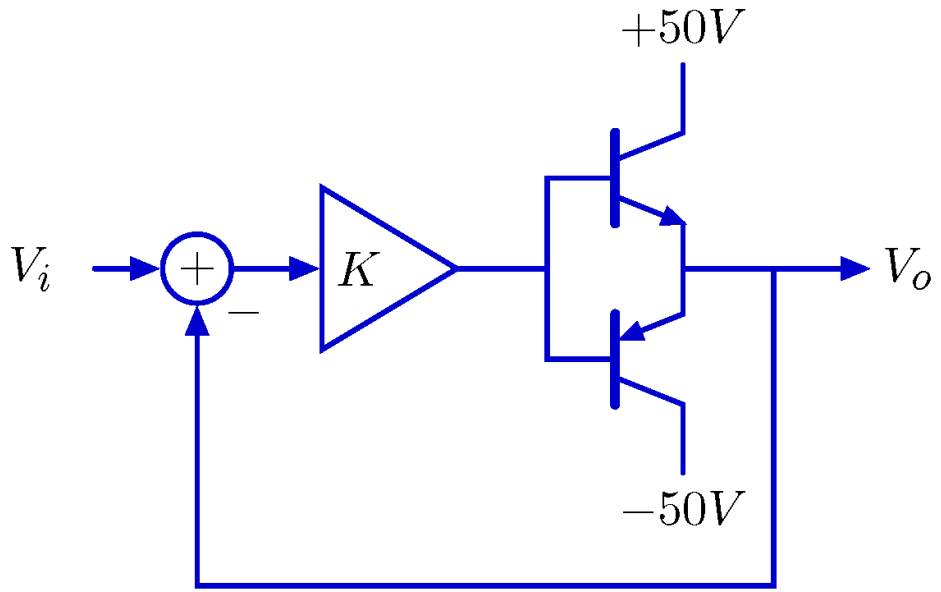
As K increases, feedback reduces crossover distortion.

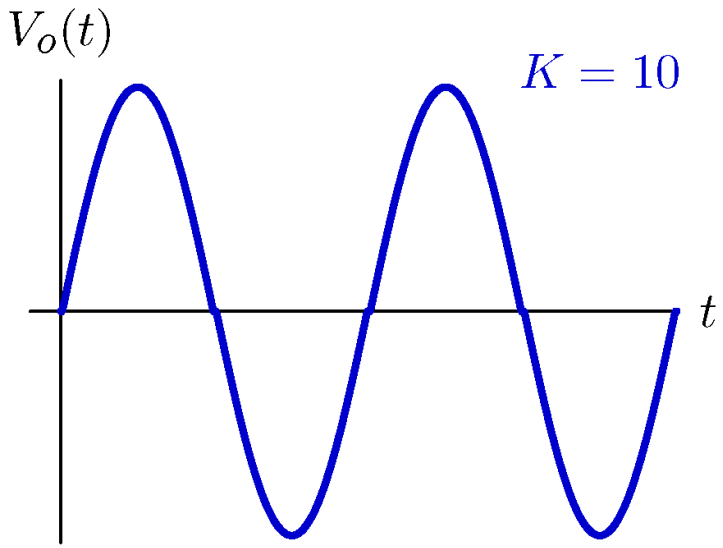




Crossover Distortion

As K increases, feedback reduces crossover distortion.





Feedback and Control

Using feedback to enhance performance.

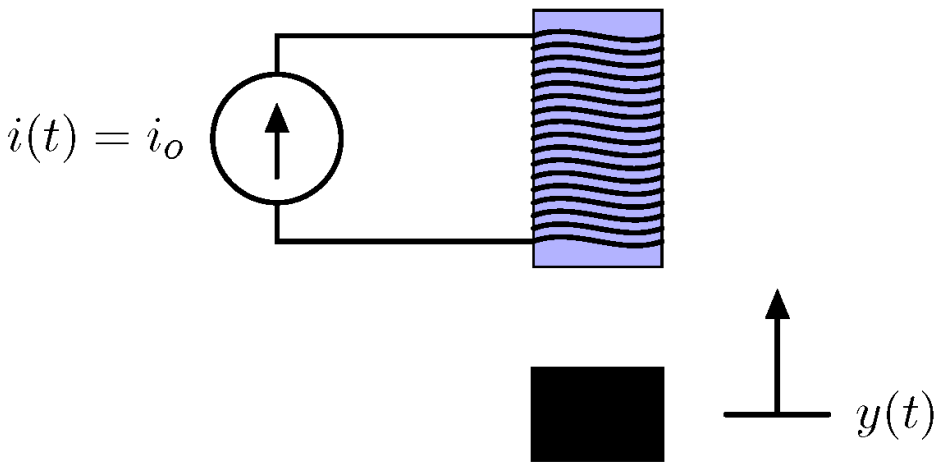
Examples:

- improve performance of an op amp circuit.
- control position of a motor.
- reduce sensitivity to unwanted parameter variation.
- reduce distortions.
- stabilize unstable systems
  - magnetic levitation
  - inverted pendulum

Control of Unstable Systems

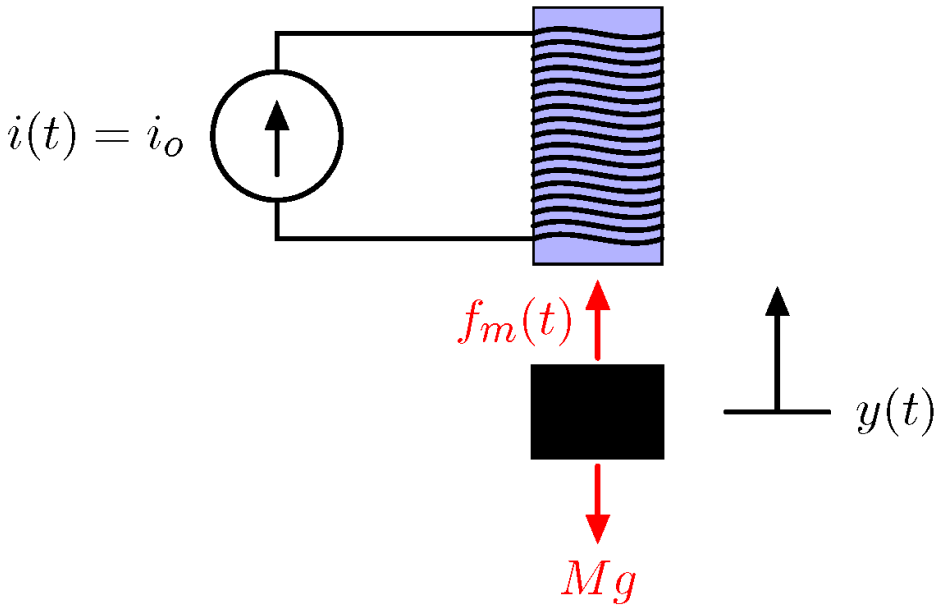
Feedback is useful for controlling **unstable** systems.

Example: Magnetic levitation.



Control of Unstable Systems

Magnetic levitation is unstable.



Equilibrium ($y = 0$): magnetic force $f_m(t)$ is equal to the weight Mg.

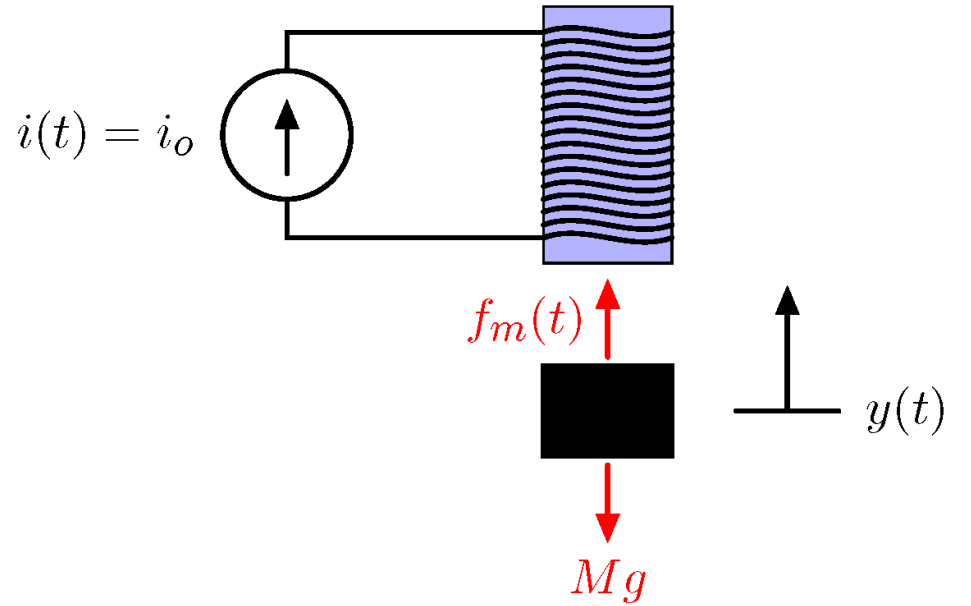
Increase $y \rightarrow$ increased force $\rightarrow$ further increases y.

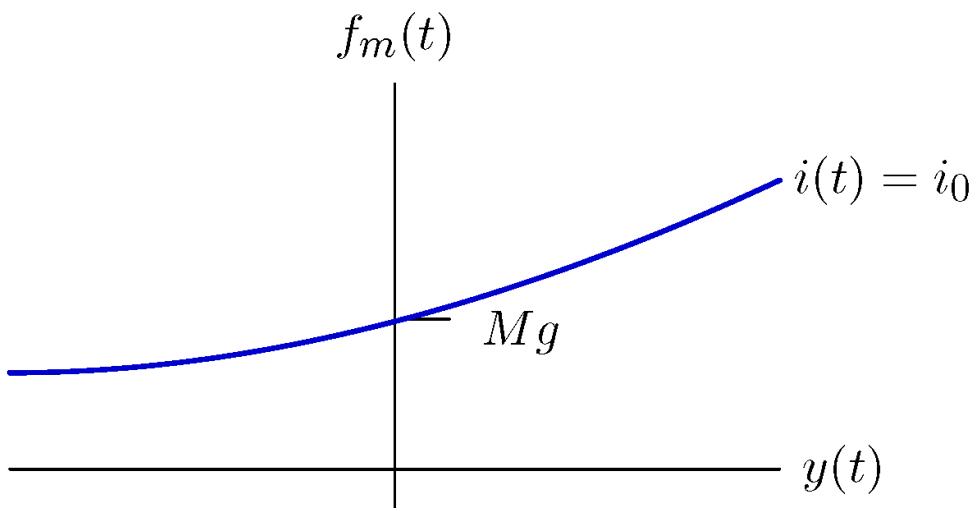
Decrease $y \rightarrow$ decreased force $\rightarrow$ further decreases y.

Positive feedback!

Modeling Magnetic Levitation

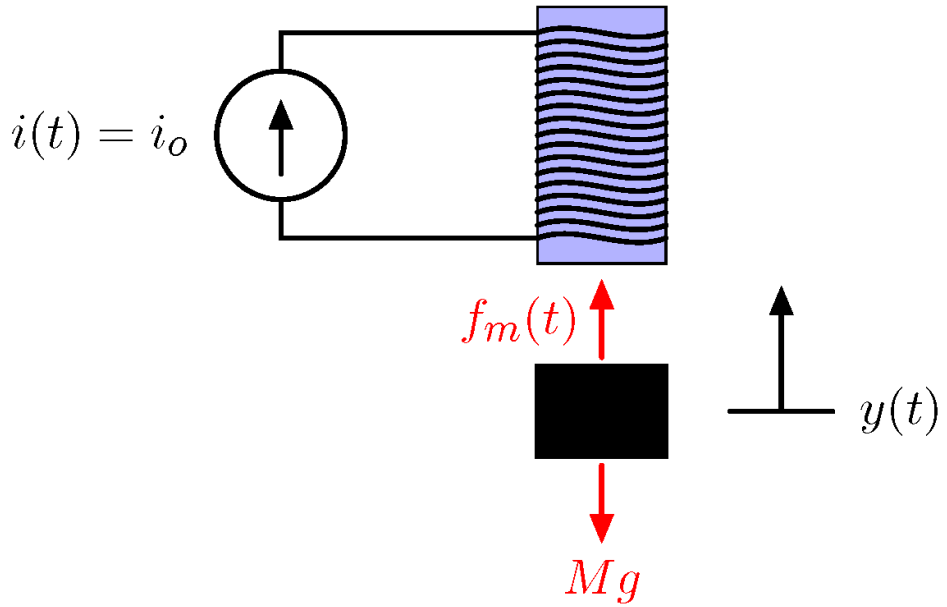
The magnet generates a force that depends on the distance $y(t)$.



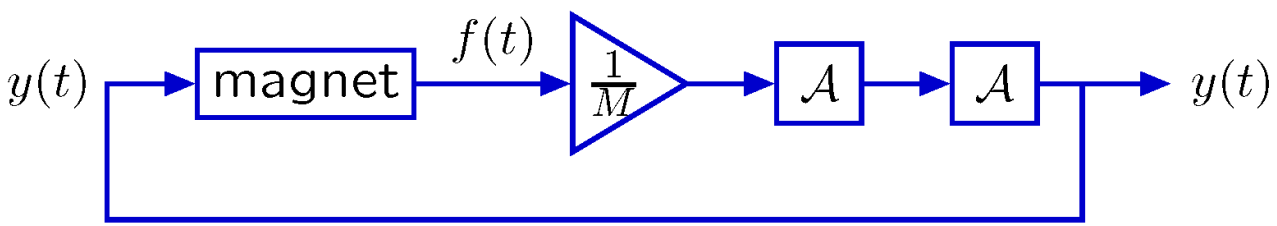


Modeling Magnetic Levitation

The net force accelerates the mass.

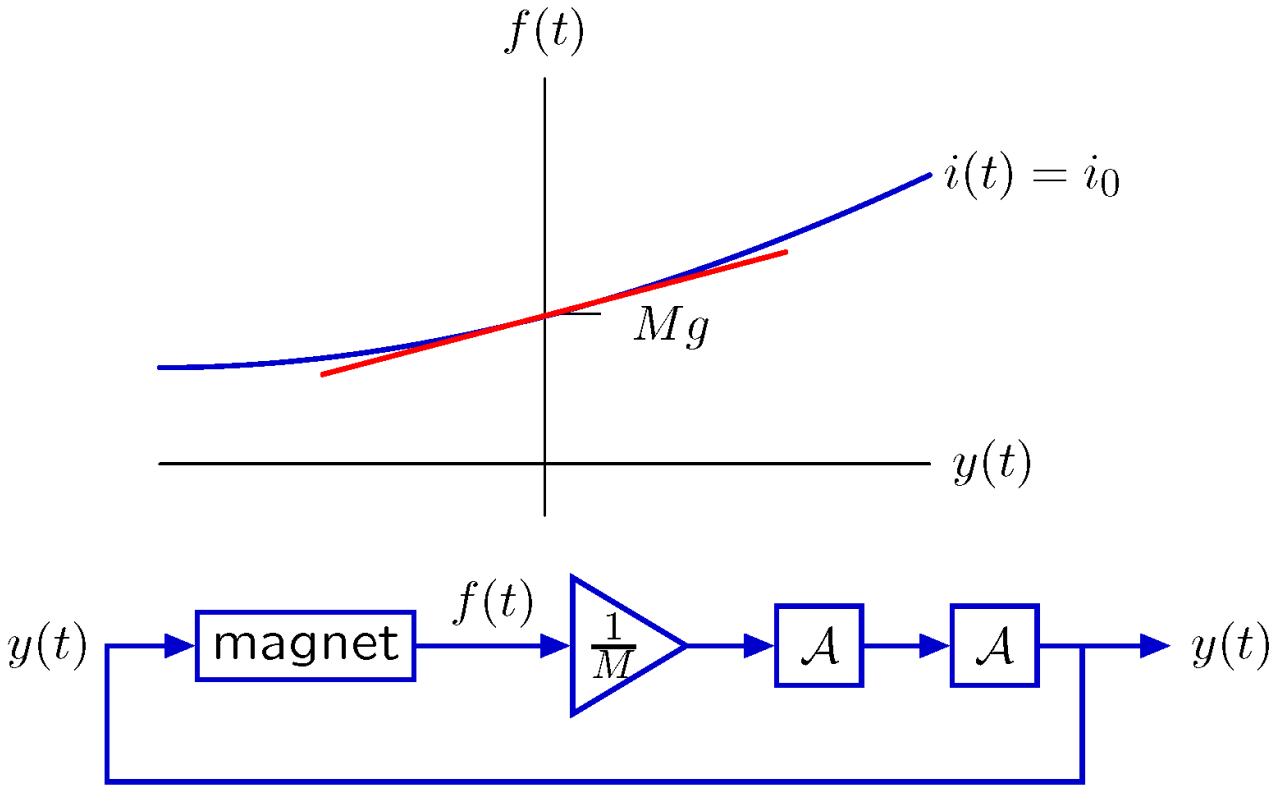


$$f_m(t) - Mg = f(t) = Ma = M\ddot{y}(t)$$



Modeling Magnetic Levitation

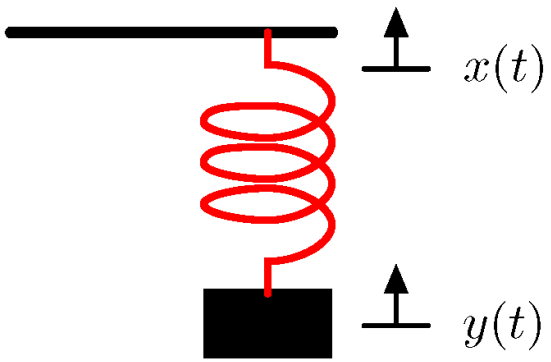
Over small distances, magnetic force grows $\approx$ linearly with distance.

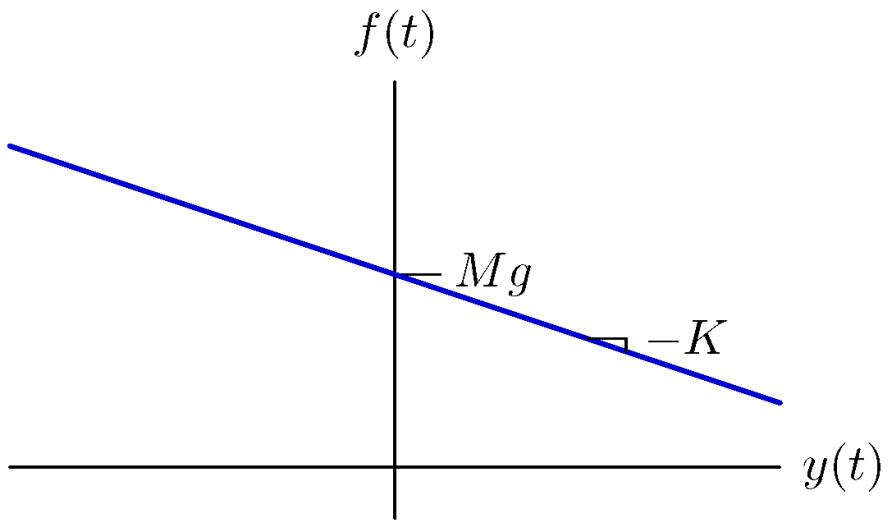


Levitation with a Spring

Relation between force and distance for a spring is opposite in sign.

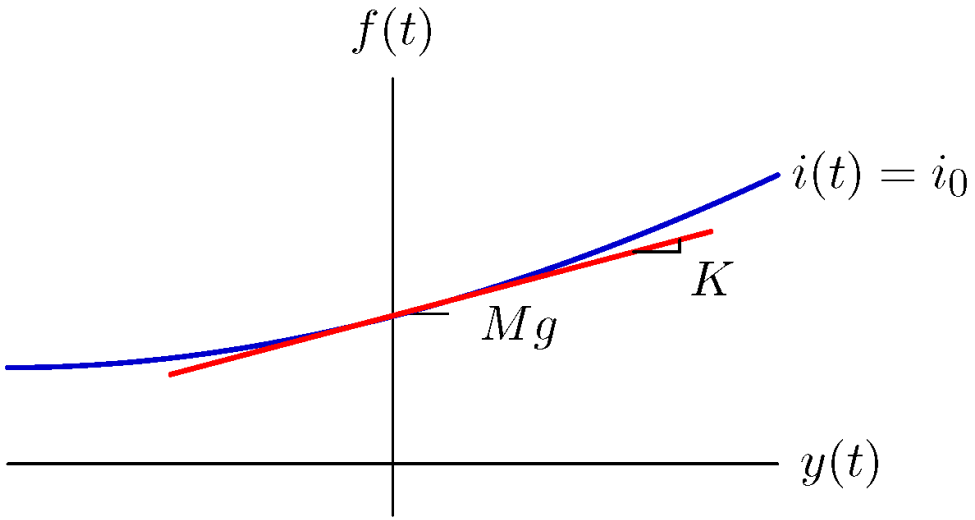
$$F = K\Big(x(t) - y(t)\Big) = M\ddot{y}(t)$$



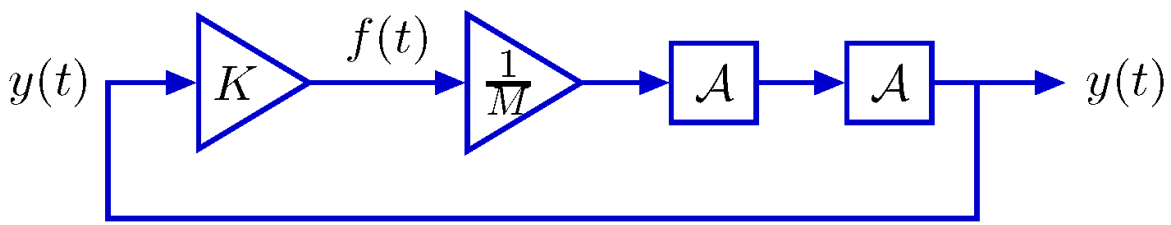


Modeling Magnetic Levitation

Over small distances, magnetic force nearly proportional to distance.



$f(t) \approx Ky(t)$

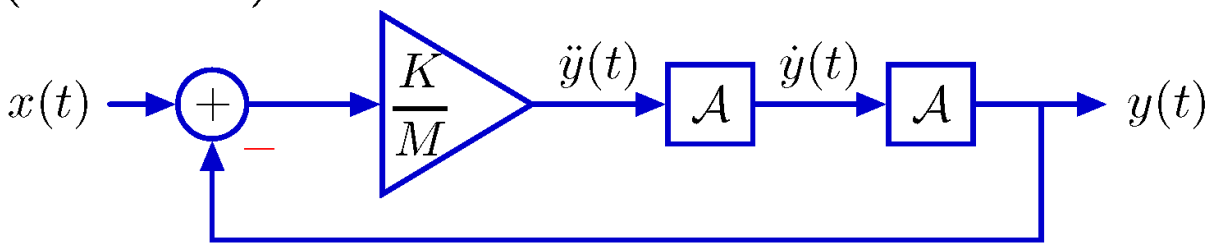


Block Diagrams

Block diagrams for magnetic levitation and spring/mass are similar.

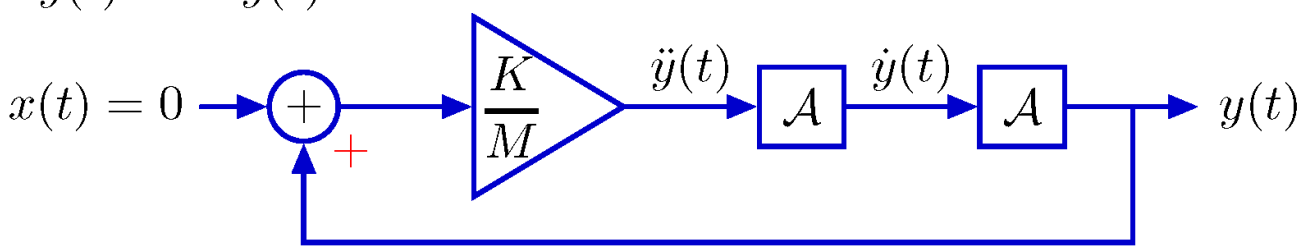
Spring and mass

$$F = K\Big(x(t) - y(t)\Big) = M\ddot{y}(t)$$



Magnetic levitation

$$F = Ky(t) = M\ddot{y}(t)$$

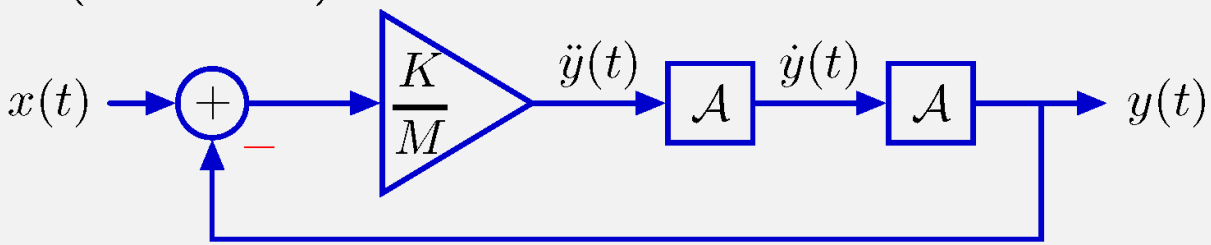


Check Yourself

How do the poles of these two systems differ?

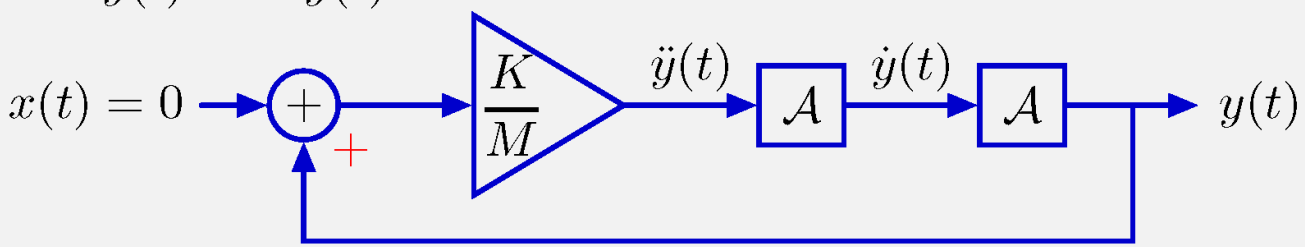
Spring and mass

$$F = K\Big(x(t) - y(t)\Big) = M\ddot{y}(t)$$



Magnetic levitation

$$F = Ky(t) = M\ddot{y}(t)$$



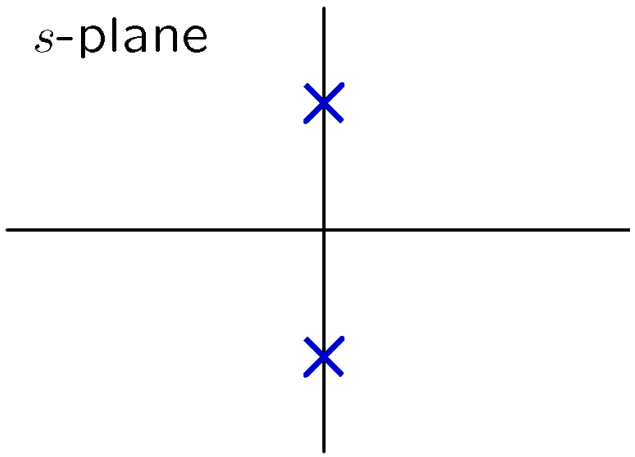
Check Yourself

How do the poles of the two systems differ?

Spring and mass

$$F = K\Big(x(t) - y(t)\Big) = M\ddot{y}(t)$$

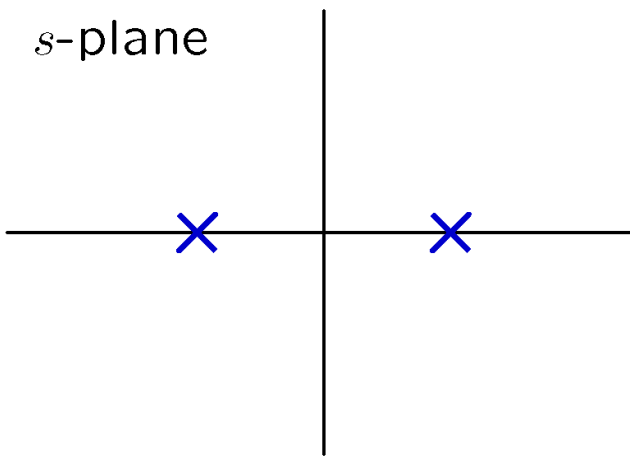
$$\frac{Y}{X} = \frac{\frac{K}{M}}{s^2 + \frac{K}{M}} \quad \rightarrow \quad s = \pm j\sqrt{\frac{K}{M}}$$



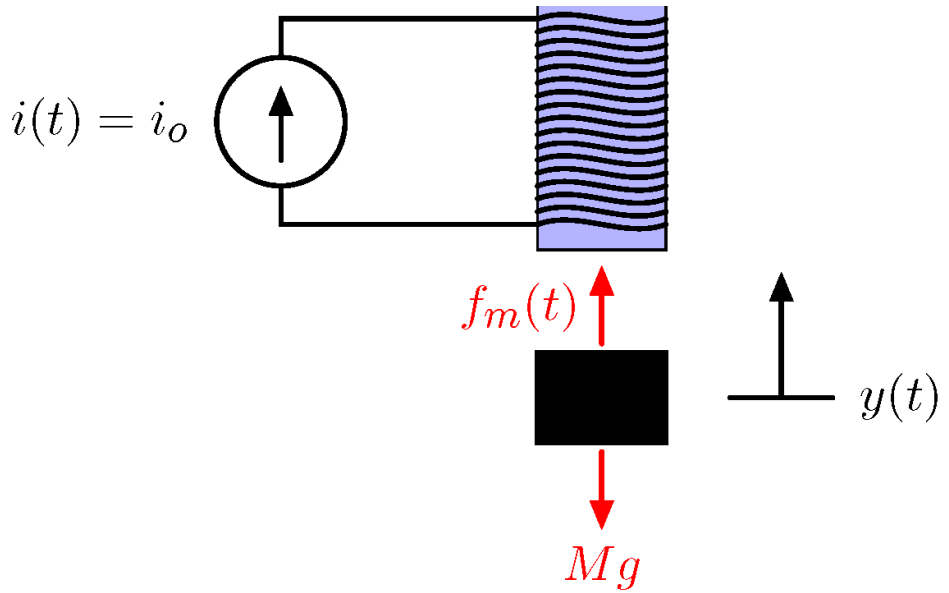
Magnetic levitation

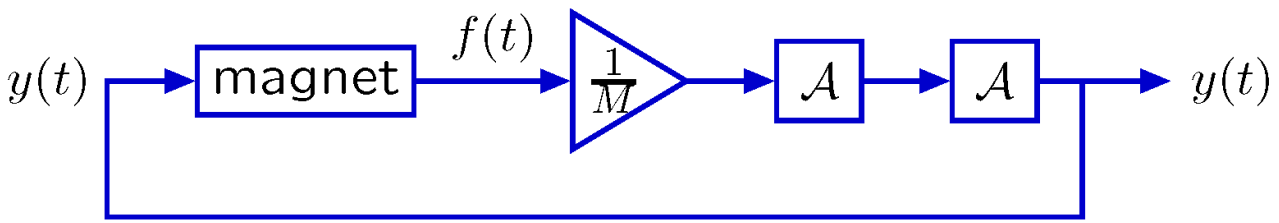
$$F = Ky(t) = M\ddot{y}(t)$$

$$s^2 = \frac{K}{M} \quad \rightarrow \quad s = \pm\sqrt{\frac{K}{M}}$$



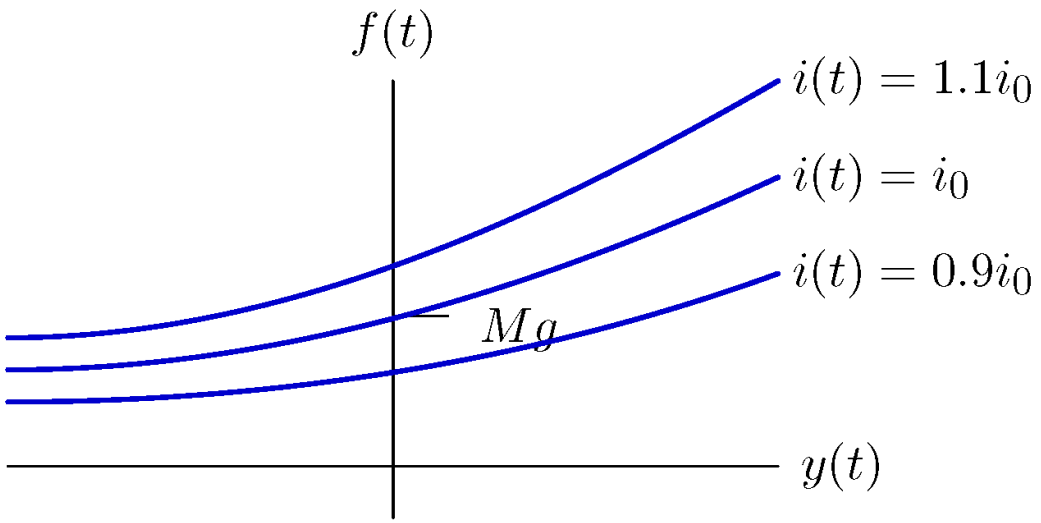
Magnetic Levitation is Unstable





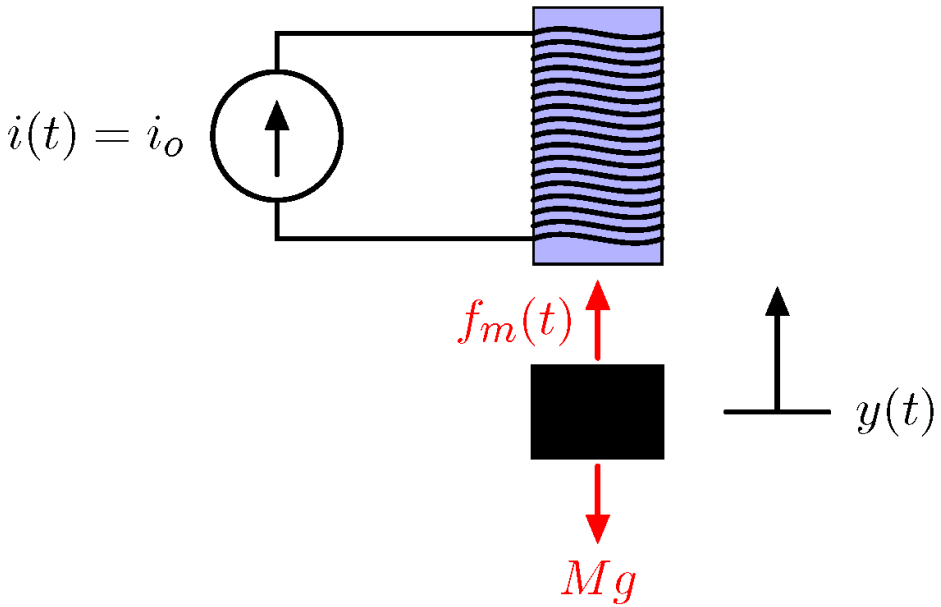
Magnetic Levitation

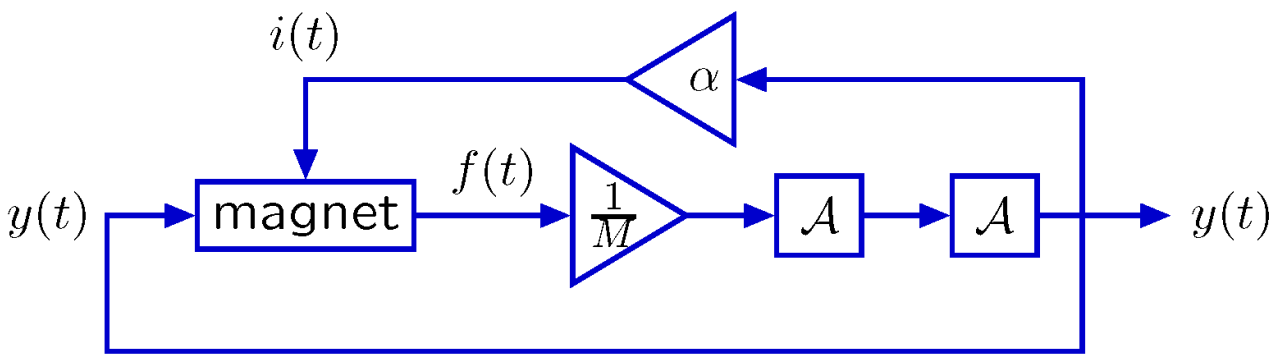
We can stabilize this system by adding an additional feedback loop to control $i(t)$.



Stabilizing Magnetic Levitation

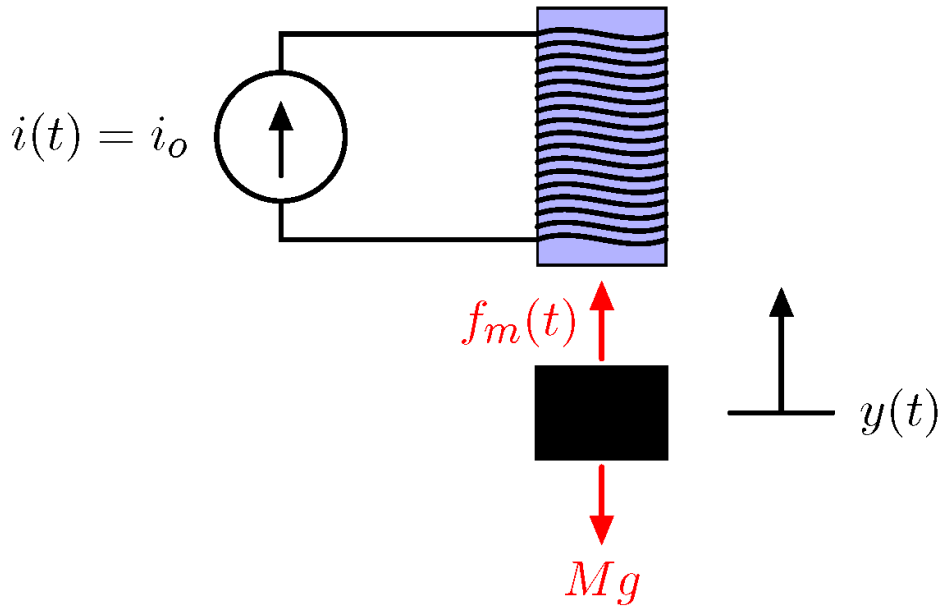
Stabilize magnetic levitation by controlling the magnet current.

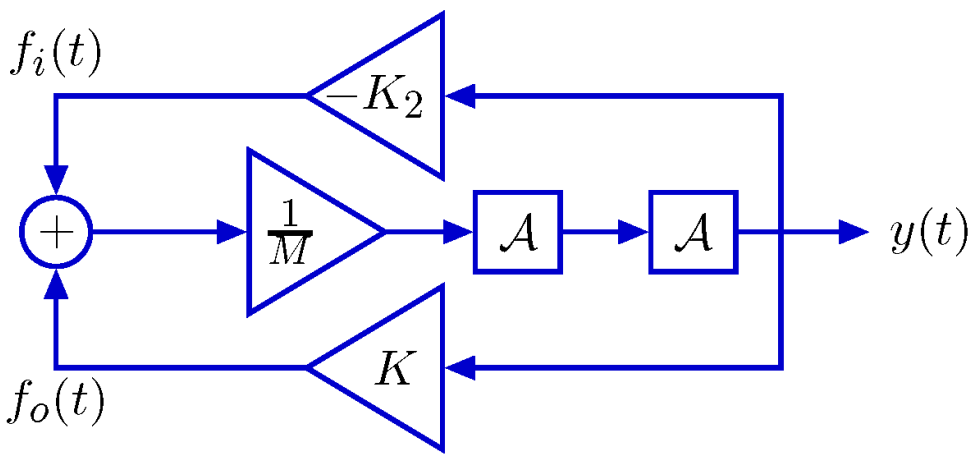




Stabilizing Magnetic Levitation

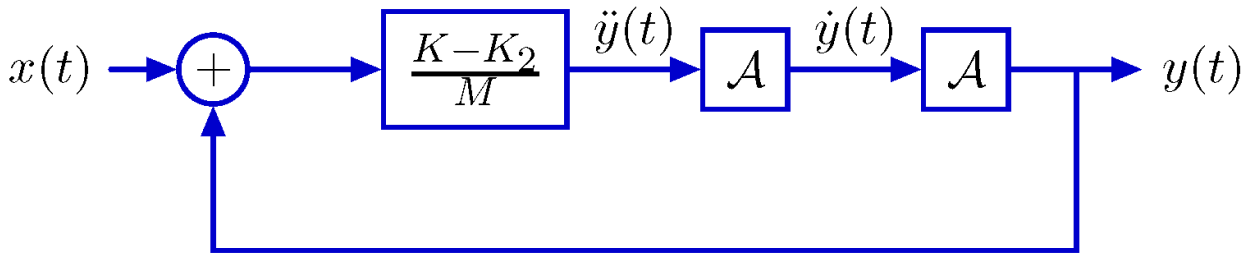
Stabilize magnetic levitation by controlling the magnet current.

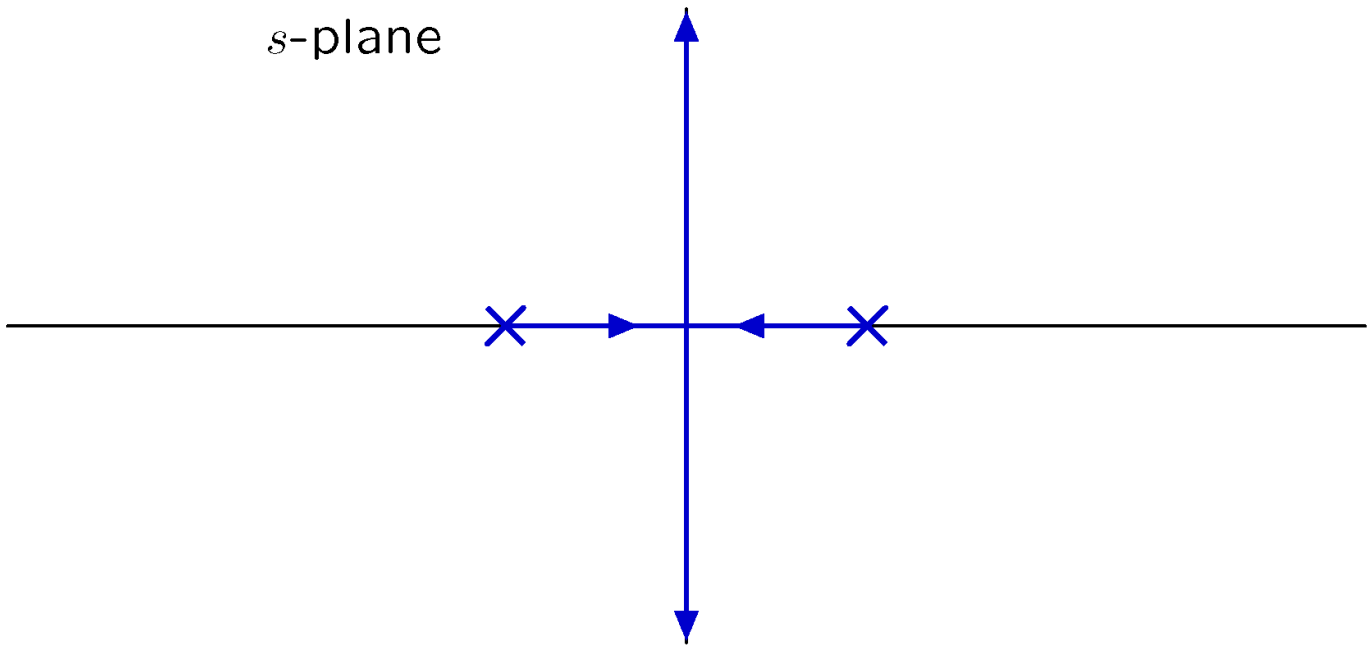




Magnetic Levitation

Increasing K_2 moves poles toward the origin and then onto $j\omega$ axis.

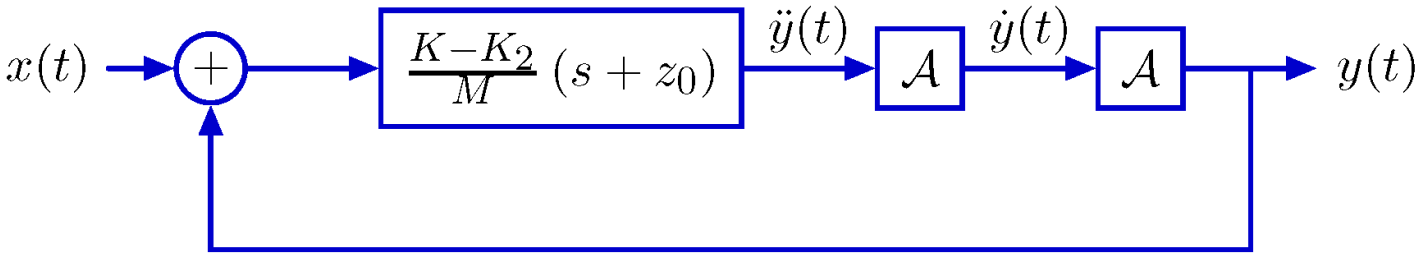


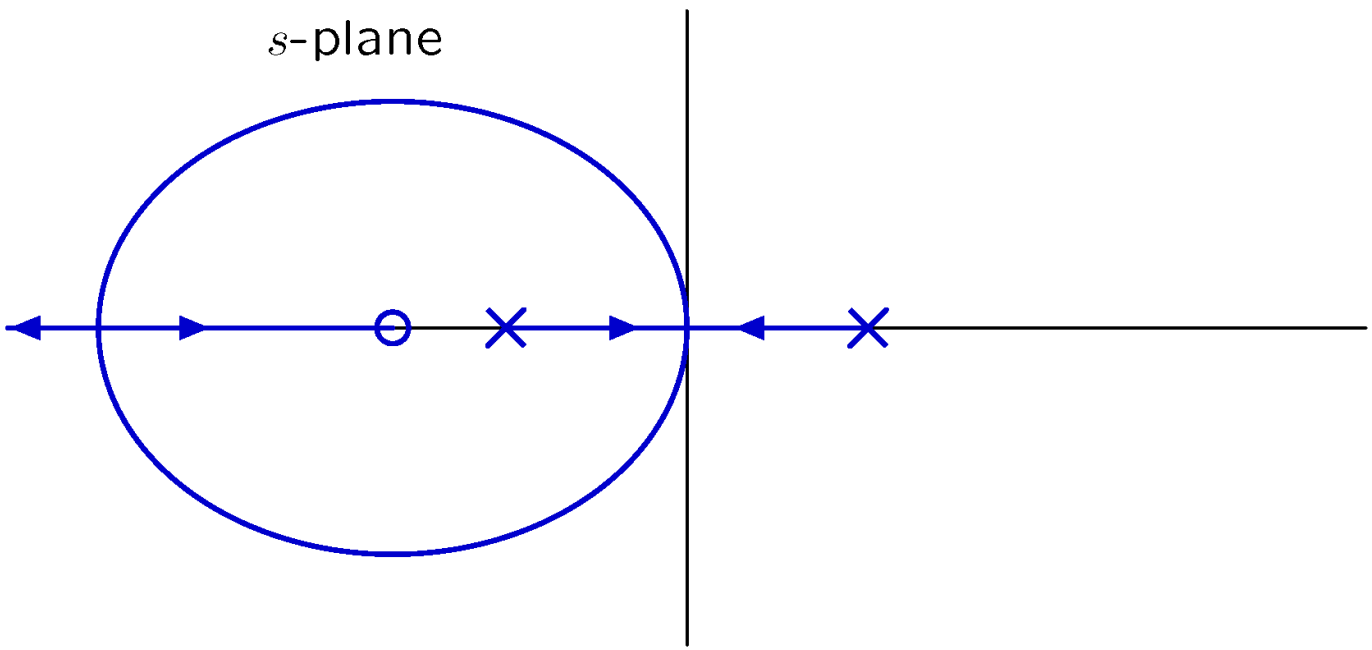


But the poles are still marginally stable.

Magnetic Levitation

Adding a zero makes the poles stable for sufficiently large K_2.

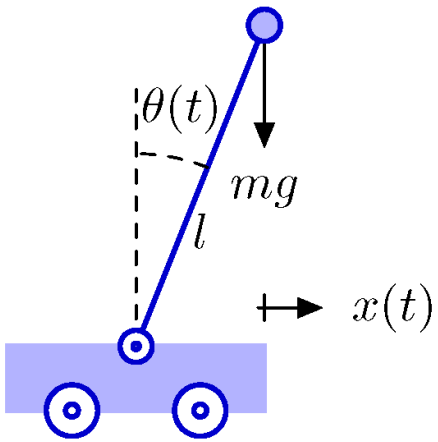




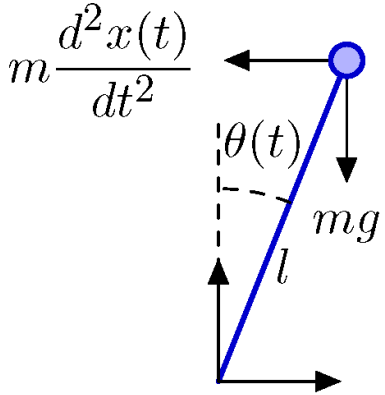
Try it: Demo [designed by Prof. James Roberge].

Inverted Pendulum

As a final example of stabilizing an unstable system, consider an inverted pendulum.



lab frame
(inertial)

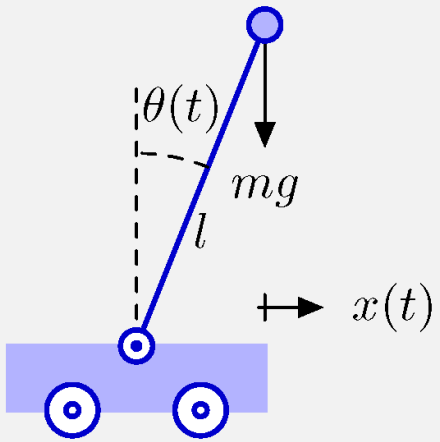


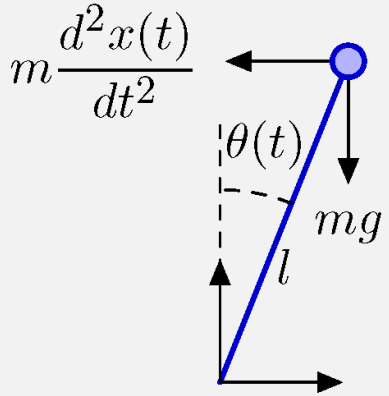
cart frame
(non-inertial)

$$\underbrace{ml^2}_{I}\frac{d^2\theta(t)}{dt^2} = \underbrace{mg}_{\text{force}}\underbrace{l\sin\theta(t)}_{\text{distance}} - \underbrace{m\frac{d^2x(t)}{dt^2}}_{\text{force}}\underbrace{l\cos\theta(t)}_{\text{distance}}$$

Check Yourself: Inverted Pendulum

Where are the poles of this system?

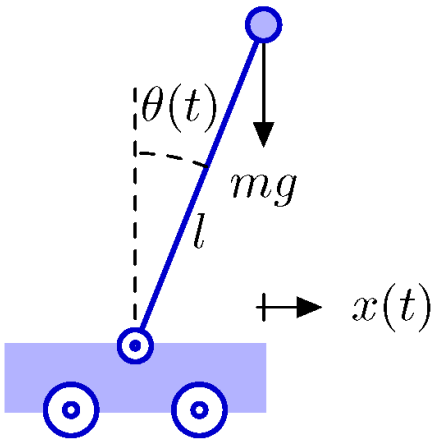


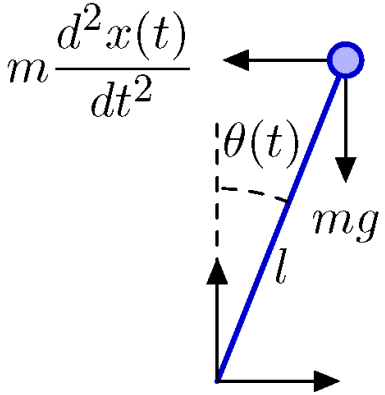


$$ml^2\frac{d^2\theta(t)}{dt^2} = mgl\sin\theta(t) - m\frac{d^2x(t)}{dt^2}l\cos\theta(t)$$

Check Yourself: Inverted Pendulum

Where are the poles of this system?





$$ml^2\frac{d^2\theta(t)}{dt^2} = mgl\sin\theta(t) - m\frac{d^2x(t)}{dt^2}l\cos\theta(t)$$

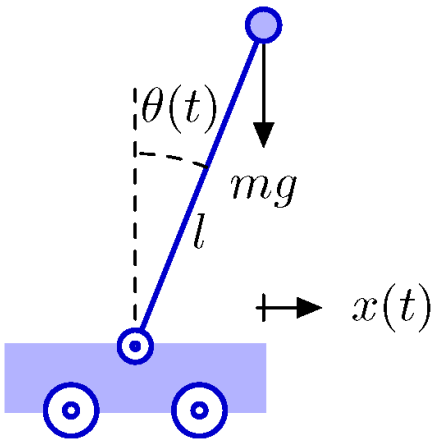
$$ml^2\frac{d^2\theta(t)}{dt^2} - mgl\theta(t) = -ml\frac{d^2x(t)}{dt^2}$$

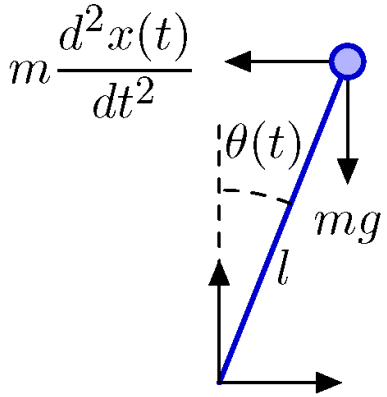
$$H(s) = \frac{\Theta}{X} = \frac{-mls^2}{ml^2s^2 - mgl} = \frac{-s^2/l}{s^2 - g/l}$$

poles at $s = \pm\sqrt{\frac{g}{l}}$

Inverted Pendulum

This unstable system can be stablized with feedback.





Try it. Demo. [originally designed by Marcel Gaudreau]

Feedback and Control

Using feedback to enhance performance.

Examples:

- improve performance of an op amp circuit.
- control position of a motor.
- reduce sensitivity to unwanted parameter variation.
- reduce distortions.
- stabilize unstable systems
  - magnetic levitation
  - inverted pendulum

Assignments

- Reading Assignment: Chap. 11.0-11.2
- Review Chap. 9.7-9.8, 10.7-10.8
- Mid-term Exam: 04/18/2024, 19:00-21:00, West 1-306